A rotary direct air capture adsorber with ultrahigh productivity for efficient CO_2 enrichment in greenhouse

Junye Wu[1, 2, 3], Kuihua Wang[1], Junde Zhao[1], Yanlin Chen[1], Zhuozhen Gan[1], Xuancan Zhu[1], Ruzhu Wang[1], Bo Wang[2, 3], Chi-Hwa Wang[2, 4], Yen Wah Tong[2, 4], and Tianshu Ge[1, 2 *]

[1] Research Center of Solar Power & Refrigeration, Institute of Refrigeration and Cryogenics, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai, 200240, China
[2] Energy and Environmental Sustainability Solutions for Megacities (E2S2), Campus for Research Excellence and Technological Enterprise (CREATE), 138602, Singapore
[3] NUS Environmental Research Institute, National University of Singapore, Singapore 138602, Singapore
[4] Department of Chemical and Biomolecular Engineering, National University of Singapore, 4 Engineering Drive 4, Singapore 117585, Singapore

[*] Correspondence: baby_wo@sjtu.edu.cn (T. G.)

SUMMARY

The continuous growth of atmospheric CO_2 concentration has motivated widespread research of negative emission technologies (NETs). Direct air capture (DAC) is a promising NET but remains under paced in deployment efforts because of high energy demand and cost. Creative implementations are required to lower the barrier to economic applicability of DAC systems. To this end, this work presents an innovative strategy to capture CO_2 from air using a rotary adsorber and delivers it to greenhouses to boost plant growth. We present a novel synthesis of a structured adsorbent with low pressure drop, good adsorption properties, and low production cost. On this basis, a rotary adsorber prototype was constructed and subjected to experiments. The results demonstrate the prototype's capability in producing 700–2000 ppm CO_2, suitable for use in greenhouses, when capturing 50–80% of CO_2 in air. The regeneration temperature was relatively low, ranging from 60–100°C, and required operation energy of only 2–4.2 kWh kg^{-1}. Additionally, the compact and efficient design of the device allows for continuous operation, eliminates the need for complex controls, and boasts a productivity of 9.1 kg h^{-1} m^{-3}(volume-based) or 0.664 g g^{-1} day^{-1} (mass-based) which are the highest values in the current experimental research. Mathematical models were developed to uncover the underlying

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operating principles and facilitate the optimization of multiple parameters. Further, an integrated system combining DAC and greenhouse was designed. The economic analysis indicates that the adsorber captures 112 t CO_2 per year with the costs of \$127–206 t^{-1}. The CO_2 enrichment results in an extra net profit of \$15350–24175 $year^{-1}$, highlighting the adsorber's potential for scaled-up implementation and application.

Keywords: direct air capture, rotary adsorber, carbon dioxide, amine, temperature swing adsorption

INTRODUCTION

Avoiding climate disasters caused by global warming is pivotal to the survival of all human beings. According to the recent report by Intergovernmental Panel on Climate Change (IPCC), a rapid decarbonization in energy sectors is likely insufficient to limit the global temperature rise well below 2°C by the end of 21st century.[1] Given the current emission rate of 40 Gt $year^{-1}$, carbon budgets will be depleted in the following 10-30 years.[2,3] Most IPCC scenarios overshoot carbon budgets at first and then remove the excess 450-1000 Gt CO_2 via negative emission technologies (NETs).[1]

There are distinct classes of NETs, such as bioenergy with carbon capture and storage, direct air capture (DAC) of CO_2 from the atmosphere by engineered chemical reactions, enhanced weathering of minerals, afforestation and reforestation, etc.[4] Compared to other options, DAC presents a number of advantages and thus is gaining increasing attention. Given the homogeneous atmospheric CO_2 concentration globally (approximately 415 ppm)[5], DAC facilities can be strategically in areas with access to low-carbon energy sources and/or proximity to pipeline infrastructure to minimize transportation costs.[6] Besides, most DAC designs do not consume major land and water resources,[4,7] relieving the food concerns about NETs.[8] There are two main DAC technologies: one is the absorption-based DAC[9-11] that relies on alkali aqueous solutions and the other is the adsorption-based DAC that captures CO_2 using amines bonded to porous solids.[12-14] Typically, absorption-based DAC requires a high temperature (800–1000°C) to be regenerated.[15] It also

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suffers from large energy consumption resulting from water evaporation[16] and equipment damage caused by alkali corrosion.[17] Overcoming these drawbacks, adsorption-based DAC can function with a low-temperature heat source (80–120°C) and has more favorable contact with air.[18-20] Techno-economic analysis and life cycle assessments have revealed its potential for large-scale deployment and capturing gigaton-level carbon emissions.[21-25]

To fulfill its role, however, DAC still needs substantial improvement and breakthrough. Recently, plenty of studies have focused on material synthesis [26-33] and shaping techniques to transform them into structured adsorbents.[34-37] For efficacious applications in scaled-up facilities, different cyclic modes including temperature swing adsorption,[38,39] moisture swing adsorption[40], and electric swing adsorption[41,42] have been developed. Based on these materials and processes, corresponding reactors have been designed for DAC, most of which are fixed beds.[43-49] Though there are commercial demonstrations of DAC systems,[15] experimental research on the device level remains scarce. The high energy consumption of 4–16 kWh kg^{-1}[46] and expensive capture cost of \$100–1000 t^{-1}[15,50,51] of current DAC processes continue to pose significant challenges to their widespread implementation. At the same time, existing DAC facilities aim to improve the capture capacity, but ignore the equipment structure complexity and footprint. Particularly, the measure of specific productivity is barely mentioned, the few studies reported the existing productivity of 0.075–0.265 g g^{-1} day^{-1}.[38,39,52] The concept of a rotating bed was proposed in an early period but has rarely been used to adsorb CO_2.[53-57] Among them, the rotary adsorption machine developed by Svante represents a notable study in carbon capture, but it has only been reported in post-combustion scenarios.[58] Rotating beds offer several advantages, including continuous operation, simple equipment, and lower energy consumption. In the context of DAC, where a large volume of air must be processed, the combination of structured adsorbent and rotating beds is expected to significantly enhance efficiency. However, this approach has received limited attention and investigation to date (a more detailed summary of DAC device studies and rotating bed is provided in the **Supplemental**

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Information).

Estimations by the United Nations suggest that the global population will reach 10.1 billion by 2050.[59] This growth rate makes meeting the future food demand a paramount issue. Controlled environment agriculture (CEA), whose representative solution is the greenhouse, is recognized as an effective strategy to solve the food scarcity problem.[60] In greenhouses, the CO_2 concentration is typically elevated to boost plant growth and increase agricultural production.[61] A conventional method for CO_2 enrichment is adding organic matter bags or fermenting buckets that release CO_2 into a greenhouse.[62] These materials are inexpensive, but they lose effectiveness over time and the amount of CO_2 delivered cannot be regulated. Alternatively, compressed CO_2 in tanks or cylinders is used to supply CO_2.[63] In this case, the delivery of CO_2 is flexible and controllable, but the high purchase cost of liquid CO_2 limits their applications. The combustion of hydrocarbon fuels is also utilized to generate CO_2.[64,65] Nevertheless, combustion without adequate oxygen may produce impurities that are harmful to plants. Seeking economic, environmental-friendly, and effective CO_2 enrichment strategy is imperative for further development of agriculture.

The integration of DAC and CO_2 enrichment presents a promising opportunity to achieve both negative emissions and improved crop yields. Inspired by the desiccant wheel device, this work proposes a new approach to capture CO_2 from air using a novel rotary adsorber and supply the concentrated CO_2 to a greenhouse. The amine-based structured adsorbent was developed with innovative features. It boasts low pressure drop, high heat/mass transfer, good adsorption properties, low production cost, and scalability. On this basis, we constructed a rotary adsorber prototype and conducted experiments. Results show that the rotary adsorber can produce 700–2000 ppm of CO_2-enriched air suitable for greenhouse use, with 50–80% capture rates. Additionally, the device has a low energy consumption of 2–4.2 kWh kg^{-1} and requires a low regeneration temperature of 60–100°C. The ingenious design of the device enables continuous operation in a compact space, eliminating the need for sophisticated control systems. This results in an impressive volumetric

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productivity of 9.1 kg m^{-3} h^{-1} or mass productivity of 0.664 g g^{-1} day^{-1}. Additionally, we have developed a DAC-greenhouse integrated system. Economic analysis demonstrates that the device has a capture capacity of 112 t CO_2 per year, a capture cost of \$127–206 t^{-1}, and the potential to generate extra annual net profit of \$15350–24175. These results highlight the potential for widespread adoption and implementation of the device.

THE ROTARY ADSORBER DESIGN

Due to the extremely low concentration of CO_2 in air, a significant challenge related to DAC is that a substantial volume of air must be processed to extract a practical amount of CO_2. Apparently, the conventional strategies that simply rely on packed beds with solid particles will not be feasible because of the poor gas-solid contact and considerable gas flow pressure drop. Therefore, we develop a structured adsorbent that can selectively capture CO_2 from the air by a coating method. In addition to desirable adsorption properties, the material has facile synthesis, low production cost, enhanced heat and mass transfer, and tunable structure. As shown in **Fig. 1** (A), a wheel-shaped honeycomb made of ceramic fiber was used as the substrate. After being attached with polyethyleneimine-functionalized silica sol (PEI-SiO_2), it was wrapped by sheet metal and installed into the adsorber device.

As shown in **Fig. 1** (B), in the device, the honeycomb is divided into two sections by partitions and flexible seals: the adsorption section and the regeneration section. The adsorption section and regeneration section are connected to the adsorption and regeneration flow paths, respectively. The ambient air is simultaneously introduced into the adsorption flow path (named “process air”) and regeneration flow path (named “regeneration air”) by fans in a countercurrent manner. The regeneration air is heated by the electric heater. Driven by the motor, the honeycomb keeps rotating and makes the solid material move between the adsorption and regeneration sections. Therefore, a continuous adsorption/desorption cycle is achieved within

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a single vessel: CO_2 is adsorbed when the process air contacts the solid material located in the adsorption section, and then gets desorbed into regeneration air in the regeneration section. A return air duct is installed to enable part of the air at the outlet of the regeneration flow path to return to the inlet and mix with the fresh air, which enhances the CO_2 enrichment. The CO_2 concentration, temperature, humidity, and gas velocity in the flow paths are monitored by sensors. The gas flow rate was determined indirectly by measuring the gas velocity. The photograph of the prototype is shown in **Fig. 1** (C), and the structural parameters of the prototype are shown in **Table S3.**

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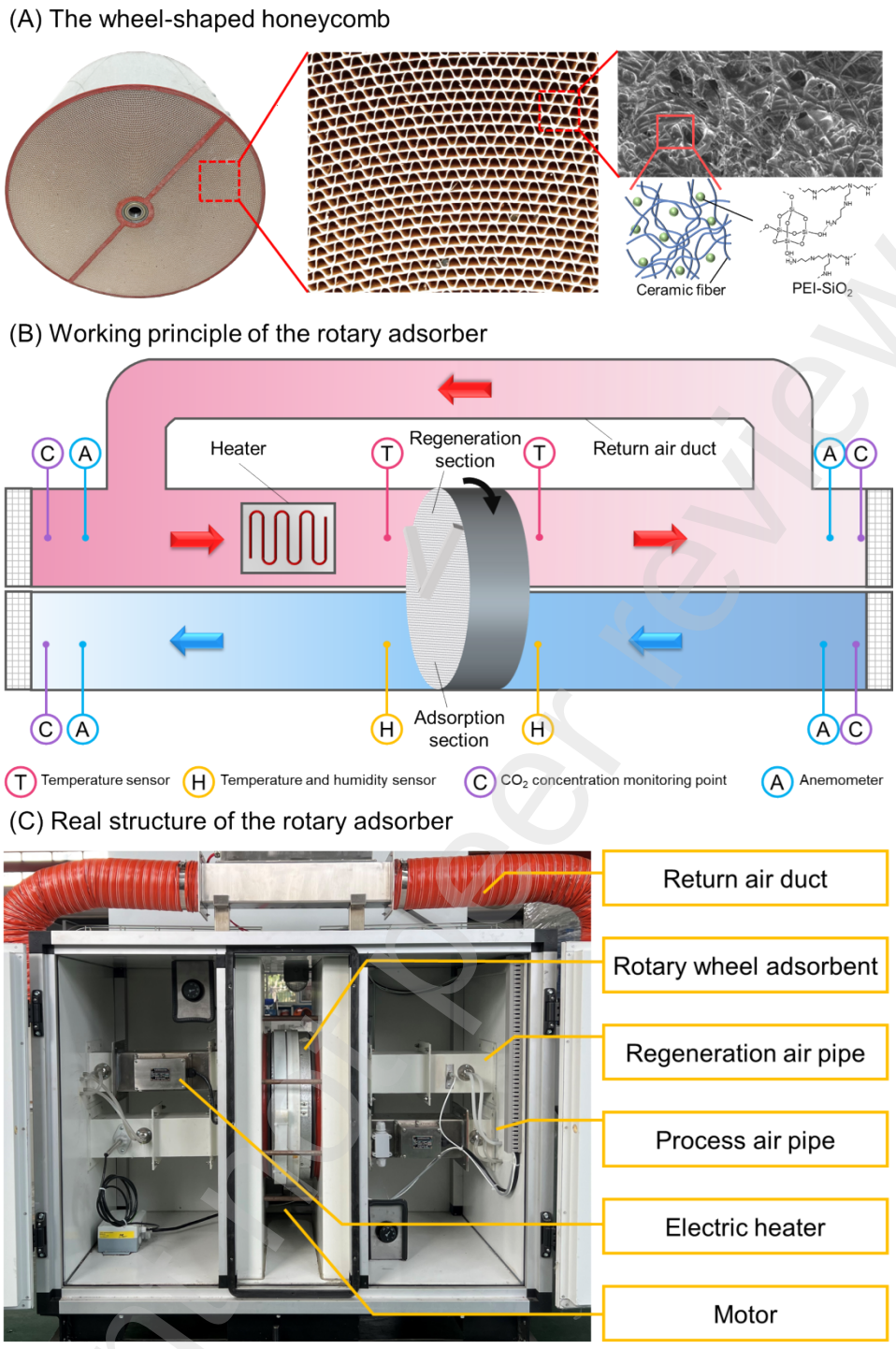


Fig. 1 The rotary adsorber design

(A) The wheel-shaped honeycomb as the structured adsorbent. The silica sol (as the support and binder) is impregnated by polyethyleneimine (as the active component), and then the polyethyleneimine-functionalized silica sol (PEI-SiO_2) is attached to ceramic fiber honeycomb (as the substrate).

(B) The diagram of the device's working principle. The wheel is divided into two sections to achieve a continuous adsorption/desorption cycle in a single vessel.

(C) The photograph of the rotary adsorber prototype. The total volume of the device is approximately 1 m^3.

RESULTS

Characterization of structured adsorbent

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Central to direct air capture technology is the use of adsorbent material. Thus, the properties of the structured adsorbent were characterized. CO_2 adsorption isotherms at different temperatures are shown in **Fig. 2** (A). Two obvious trends for uptake increase were observed in an isotherm. The first sharp uptake increase occurred at low pressures because of amine's extremely high affinity for CO_2,[32] indicating chemical adsorption dominated in this stage. The second main uptake increase occurred when pressure was greater than 500 mbar. Because ceramic fiber and silica are both porous materials, the physical adsorption was strengthened at high pressures and led to the uptake increase.[66] With the temperature increase, the isotherm shifted to the right. The maximum uptake increased at first when the temperature was lower than 75°C and then it decreased. This is because the adsorption by amine-modified materials results from the combined effects of thermodynamics and kinetics.[67] Note that the adsorption capacity is calculated per unit mass of the composite (i.e., inert honeycomb substrate was included), so it was lower than that of powdered materials. The isotherms can be well fitted by the dual-site Langmuir-Freundlich model and dual-site Toth model (**Fig. S1**).[32] Based on the fitted results, the isosteric heat of adsorption was calculated by the Clausius-Clapeyron equation.[29] It is shown in **Fig. 2** (B) that the structured adsorbent had adsorption heats of 55–70 kJ mol^{-1}, which is typical for amine-modified materials.[68]

The adsorption curves of materials test using thermal gravimetric analyzer (TGA) at different temperatures are shown in **Fig. 2** (C). The uptake increased when the adsorption temperature increased, which is consistent with the isotherms. The material exhibited fast kinetics, reaching more than 80% of the saturation capacity within 30 minutes. It is shown in **Fig. 2** (D) that the desorption rate was much faster than the adsorption rate. More than 90% of the adsorbed species could be desorbed within ten minutes when the desorption temperature was higher than 80°C. This indicates that the regeneration time can be shorter than the adsorption time. The H_2O adsorption and desorption isotherms at 25°C are shown in **Fig. 2** (E). The structured adsorbent had moderate H_2O uptake at a relative pressure of 0.3–0.7. Because of the capillary condensation,

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an adsorption–desorption hysteresis loop was observed.[69] The presence of water typically affects the CO_2 adsorption,[70] so the CO_2 uptake was measured under different relative humidity using the self-made fixed bed. The breakthrough curves are shown in **Fig. S2** and the calculated capacity is shown in **Fig. 2** (F). The CO_2 uptake increased when the atmosphere was humid, and the relative humidity (RH) of 70% led to the highest CO_2 uptake. Despite its relatively lower capacity per unit mass, the use of structured adsorbent over conventional particle solids has obvious advantages. The structured adsorbents reduce the gas pressure drop and enhance heat and mass transfer, which results in improved gas flow behavior and fast temperature swing.

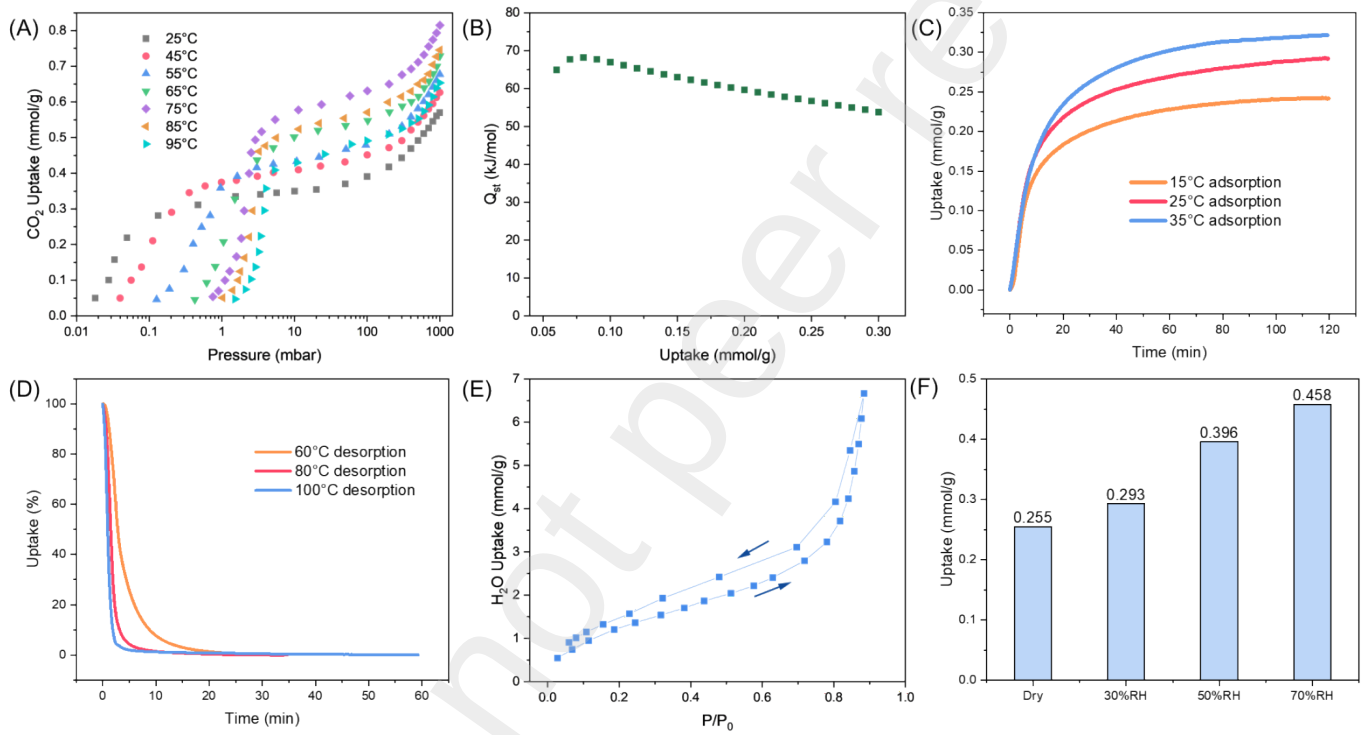


Fig. 2 The adsorption properties of structured adsorbent

(A) The CO_2 adsorption isotherms at different temperatures.

(B) The isosteric heat of adsorption calculated from CO_2 adsorption isotherms.

(C) The adsorption curves of structured adsorbent in 400 ppm CO_2 at different temperatures.

(D) The desorption curves of structured adsorbent at different temperatures.

(E) The H_2O adsorption isotherms of structured adsorbent at 25°C.

(F) The CO_2 uptake of structured adsorbent in 400 ppm CO_2 with different relative humidity at 25°C.

Experimental tests of device

The material characterization results prove the capability of structured adsorbent to capture CO_2 from air. On this basis, the device performance was examined. The unsteady-state performance of the device upon

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initial startup is described in the **Supplemental Information**. The temperature and concentration at inlet and outlet of flow paths in steady state are shown in **Fig. 3** (A-D). When the regeneration temperature was set to 70°C, the carbon capture performance of device under the condition without return air (WOR) and with return air (WR) was evidently different. It is shown in **Fig. 3** (A) that the inlet temperature in regeneration flow path was the set value 70°C, while the outlet temperature in regeneration flow path of WOR and WR was 55 and 50°C, respectively. It is shown in **Fig. 3** (B) that the inlet temperature in adsorption flow path was around 35°C, while the outlet temperature in adsorption flow path of WOR and WR was 54 and 49°C, respectively. The lower outlet temperature of WR is possibly because the longer gas flow path of WR resulted in larger heat loss. As shown in **Fig. 3** (C), given the ambient CO_2 concentration of ~430 ppm, the outlet concentration in adsorption flow path of WOR and WR was 120 and 150 ppm, respectively. As shown in **Fig. 3** (D), compared to the outlet concentration of ~700 ppm in regeneration flow path of WOR, the outlet concentration of WR was much higher (~1200 ppm). This indicates that the return air strategy is effective in improving the CO_2 concentration in product stream. The higher outlet CO_2 concentration of the adsorption flow path for WR might be due to the higher CO_2 concentration on the regeneration side, which led to inadequate regeneration of the adsorbent. This, in turn, decreased the carbon capture rate. Note that compared to WOR, greater fluctuations in temperature and concentration were observed for WR. This is because the air flow was relatively unstable when it went through a longer path and encountered larger gas flow resistance.

When return air is used, variation in regeneration temperature can affect regeneration degree of adsorbent, and thus the carbon capture performance of device. When regeneration temperature was set to 60, 70, 80 and 100°C, the temperature at inlet and outlet of regeneration flow path is shown in **Fig. 3** (A). The inlet temperature was basically stable at the set value, while the outlet temperature was 46, 50, 58 and 63°C, respectively. When the regeneration temperature increased, the temperature drop on the regeneration side became greater. This is because the greater temperature difference resulted in more heat transfer losses. The

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temperature at inlet and outlet of adsorption flow path is shown in **Fig. 3** (B). The inlet temperature was basically stable at 35°C, while the outlet temperature was 44, 50, 54 and 60°C, respectively. The temperature drop on the regeneration side and the temperature rise on the process side were due to the joint effects of sensible heat and adsorption heat. The outlet temperature of regeneration and adsorption flow path were close, which hints that a relatively intensive heat exchange occurred in the wheel. It is demonstrated that relatively low temperatures (60–100°C) are sufficient to regenerate the adsorbent, indicating that low grade heat source can be used to drive the device and wide application ranges are possible.

The carbon capture rate, product stream average CO_2 concentration, productivity and energy consumption of device were calculated according to the recorded CO_2 concentration, temperature and gas flow rate (detailed in **Supplemental Information**), which are shown in **Fig. 3** (E-F). It is shown in **Fig. 3** (E) that compared to WOR, the carbon capture rate decreased slightly but the product stream CO_2 concentration increased by nearly two times for WR. Meanwhile, the productivity increased and specific energy consumption decreased. As for WR regenerated at different temperatures, when regeneration temperature increased, the carbon capture rate, product stream CO_2 concentration, and productivity all increased. The specific energy consumption decreased at first and then increased. The higher temperature enabled fuller regeneration, thus the overall carbon capture performance of device increased. However, when the regeneration temperature increased, the energy input also increased and heat loss during heat exchange became more prominent. Thus, the specific energy consumption increased, which indicates a suitable regeneration temperature of ~80°C under this condition. Note that moderate temperatures (60–100°C) could regenerate the adsorbent. This advantage is assumed to be imparted by the enhanced heat and mass transfer of structured adsorbent, which is similar to results of desiccant wheels.[71,72] Additionally, the productivity reached a high value of 9.1 kg h^{-1} m^{-3} when the regeneration temperature was set at 100°C. This high performance can be attributed to the compact design of the gas-solid contactor and the rapid adsorption cycle.

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The rotary adsorber device has various adjustable parameters, most of which may affect its carbon capture performance. This gives it the flexibility to adapt to different working conditions.[73,74] Herein, the wheel rotation speed and process air flow rate were also altered to investigate their effects. The results of varying wheel rotation speed are shown in **Fig. S4**. It is shown in **Fig. S4** (A-B) that the outlet temperature in regeneration flow path decreased with rotation speed, while the outlet temperature in adsorption flow path increased. This is probably because the heat transfer became more intense when rotation speed increased. The CO_2 concentration at outlet of flow paths is shown in **Fig. S4** (C-D). With rotation speed increase, the outlet concentration of adsorption and regeneration flow path decreased and increased respectively. This indicates the capture performance of device increased, which is also shown in **Fig. S4** (E-F). This is possibly because when the rotation speed was slow, the adsorbent reached saturation but could not enter the desorption section to be regenerated. In this case, increasing rotation speed is beneficial for carbon capture. The results of varying process air flow rate are shown in **Fig. S5**. It is shown in **Fig. S5** (A-B) that with increase in process air flow rate, the outlet temperature of the regeneration and adsorption flow paths both decreased. This is possibly because the larger flow rate of air enhanced the cooling of wheel. It is shown in **Fig. S5** (C-D) that outlet CO_2 concentration of adsorption and regeneration flow paths both increased with process air flow rate, which corresponds to decreased capture rate and increased product stream average CO_2 concentration as shown in **Fig. S5** (E). This is because a larger flow rate of process air made it more difficult for adsorbent to completely capture the CO_2 in the stream. However, the overall capture capacity increased as show in **Fig. S5** (F), so the amount of CO_2 delivered to the regeneration side increased. The parametric study described above provided insight into the device's operating pattern and helped identify the suitable working conditions.

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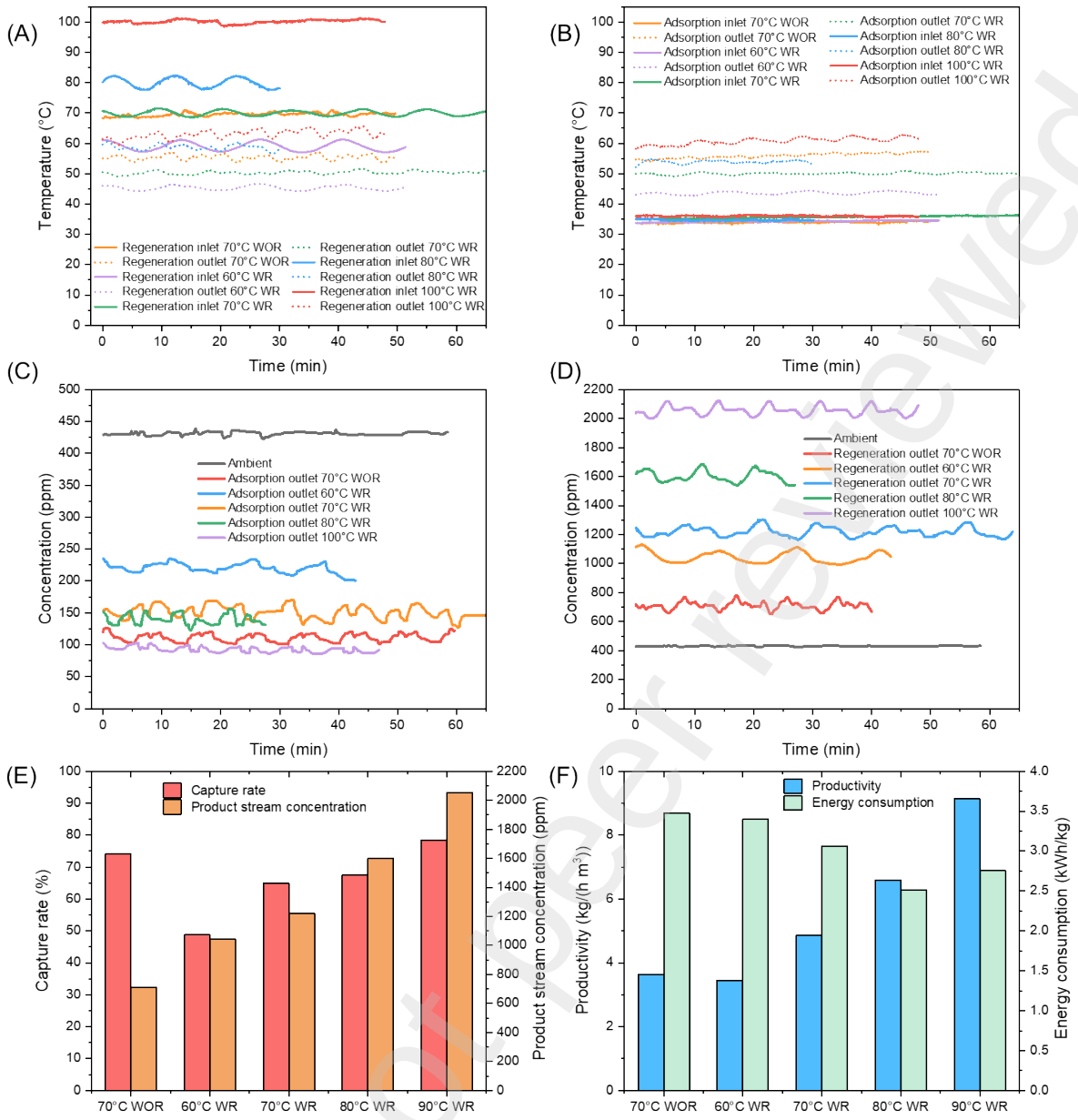


Fig. 3 The experimental tests of device

(A) The temperature in inlet and outlet of desorption flow path.

(B) The temperature in inlet and outlet of adsorption flow path.

(C) The CO_2 concentration in ambience and outlet of adsorption flow path.

(D) The CO_2 concentration in ambience and outlet of desorption flow path.

(E) The carbon capture rate and product stream CO_2 concentration of device.

(F) The productivity and energy consumption of device.

Model Analysis

Having a comprehensive understanding of the heat and mass transfer in carbon capture is instructive to improve the device performance. From the experimental point of view, it is difficult to measure thermodynamic changes of adsorbent during cycles. Therefore, a mathematical model is developed to describe

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the gas and solid behavior inside channels of honeycomb as shown in **Fig. 4** (A). The model specification is provided in the **Supplemental Information**. The input parameters in the model are shown in **Table S4-S5**. The output parameters are the temperature and CO_2 concentration in the gas and solid phase and CO_2 uptake of adsorbent. By analyzing the variation of parameters in the control volume channel, the macro performance of device can be elucidated.

For the simplicity of the process, the simulations were performed without the use of return air. The model validation was performed using the experimental data. Under same conditions, the comparison between experimental and simulated outlet temperature and CO_2 concentration is shown in **Fig. 4** (B-C). The maximum difference was smaller than 20%. The error might be caused by the neglect of the effects of H_2O adsorption and the limitations in sensor accuracy. The operation from unsteady to steady state is shown in **Fig. 4** (D). Originally, the adsorbent was in equilibrium under the atmospheric conditions. When the device started working, it experienced unsteady state in the first several cycles. The outlet CO_2 concentration differed in a series of consecutive cycles, then the difference gradually vanished and the steady state was reached.

The outlet CO_2 concentration, temperature, and CO_2 uptake of adsorbent in steady state are shown in **Fig. 4** (E-G). It is shown in **Fig. 4** (E) that when the adsorbent entered the regeneration section, the hot regeneration air purged the adsorbent and raised its temperature to nearly 80°C. At the same time, the CO_2 was released from adsorbent and the concentration in gas phase rapidly increased as shown in **Fig. 4** (F). Moreover, the desorption driving force $q_e - q$ decreased, so the desorption rate was reduced as shown in **Fig. 4** (G). Such process continued until the adsorbent entered the adsorption section. As the adsorbent entered the adsorption section, an opposite pattern was observed. Under the purge of cool process air, the adsorbent temperature and CO_2 concentration decreased sharply as shown in **Fig. 4** (E-F). At rotating degree of 178°, the CO_2 concentration reached the minimum. This was also the point where the temperature started to plateau. As shown in **Fig. 4** (G), the CO_2 uptake kept decreasing for a short period after entering the

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adsorption section. This is because the temperature was still high and the adsorbent kept desorbing CO_2. At rotating degree of 148°, the uptake reached the minimum and the adsorption process began. As the uptake increased, the adsorption rate decreased because of the reduced driving force $q_e - q$. Then the adsorbent entered the regeneration section again and the next cycle started.

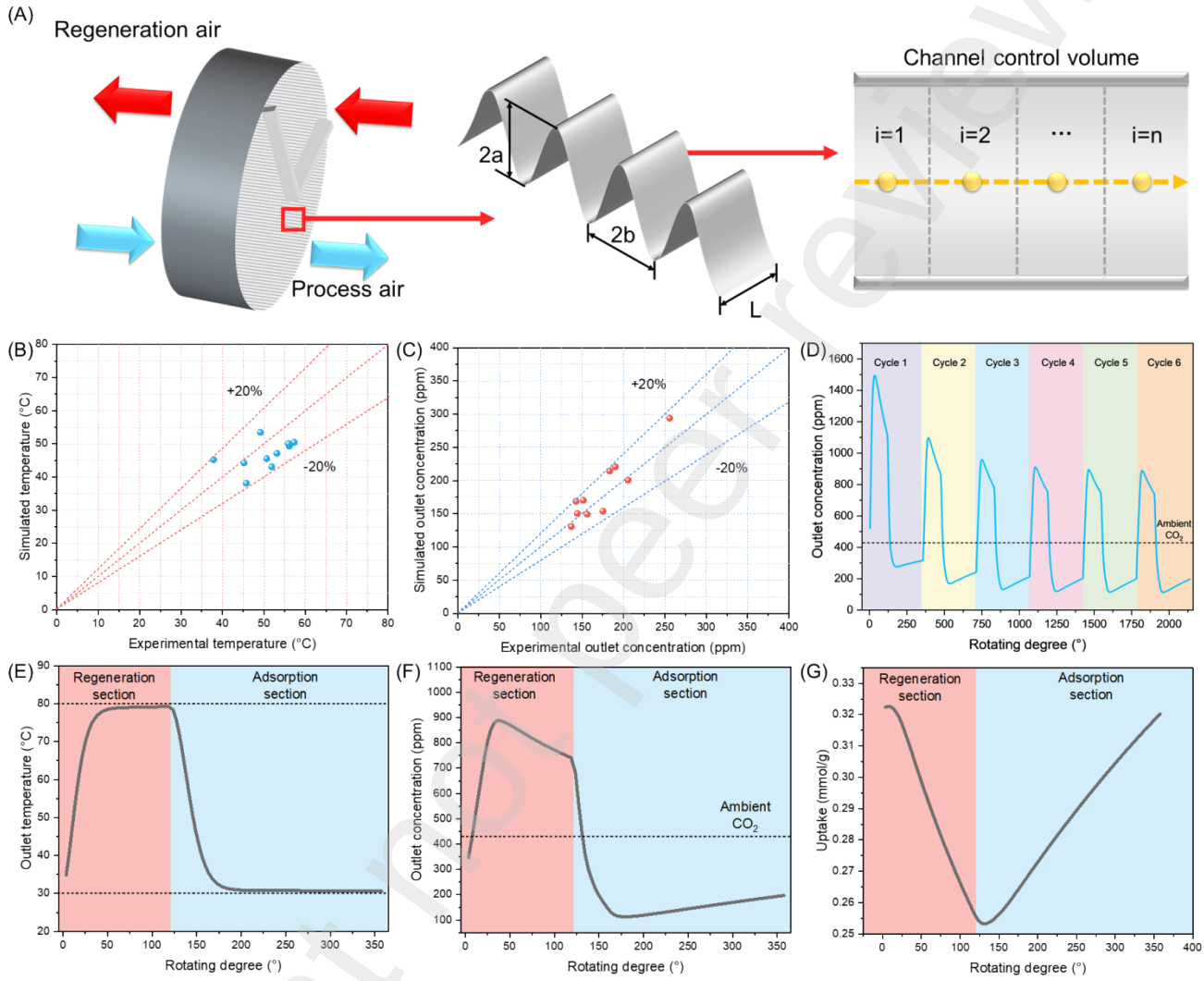


Fig. 4 The model analysis of capture process

(A) The model diagram, from left to right is the division of the wheel, the size of the sinusoidal channel and the channel control volume in the model, respectively.

(B) The comparison between experimental and simulated outlet temperature.

(C) The comparison between experimental and simulated outlet CO_2 concentration.

(D) The CO_2 concentration profile on the wheel outlet surface in continuous cycles.

(E) The CO_2 concentration profile on the wheel outlet surface in a steady state cycle.

(F) The temperature profile on the wheel outlet surface in a steady state cycle.

(G) The adsorbent CO_2 uptake profile on the wheel outlet surface in a steady state cycle.

The gas-solid interaction inside the channel was elucidated by simulating the axial parameter variation. The time in the regeneration section was divided into ten equal parts t_1, t_2, …, and t_{10}. The temperature, CO_2

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concentration, and CO_2 uptake as a function of the distance to the channel inlet and time are shown in **Fig. 5**. It is shown in **Fig. 5** (A) that the temperature decreased from the inlet to outlet. The temperature gradient from inlet to outlet was caused by the heat capacity of material and the desorption heat. The adsorbent at the inlet was the closest to hot air, so its temperature was the highest. As the hot air proceeded, the temperature in subsequent locations decreased. Furthermore, the desorption reaction is endothermic, causing a decrease in the thermal energy of the adsorbent during the desorption process, especially for the adsorbent located at the back end of the channel where the degree of desorption was higher. At t_1, the temperature experienced a greater change with a temperature difference of 12°C between the inlet and outlet. After t_2, the temperature variation became much smaller, which indicates that the heat transfer could reach equilibrium quickly.

The CO_2 concentration from the inlet to outlet are shown in **Fig. 5** (B). At t_1, the concentration first increased, then it entered a platform and decreased, which possibly resulted from the decreased temperature at the back end of channel. After t_2, the concentration kept increasing from inlet to outlet. From t_4 to t_{10}, the overall concentration inside the channel decreased. This is because there was gradually less CO_2 released as regeneration proceeded. It is shown in **Fig. 5** (C) that the CO_2 uptake of adsorbent at the back end of channel was larger than that at the front end. This is because the temperature was higher and the adsorbent was more fully regenerated at the front end. The desorption degree became larger from t_1 to t_{10}, so the overall CO_2 uptake inside the channel decreased. As shown in **Fig. 5** (D-F), the parameter variation in the adsorption section was basically opposite to that in the regeneration section. Note that the adsorption time was twice as long as the regeneration time. As a result, the interval between t_1, t_2, …, and t_{10} in adsorption section represented a longer period and the parameter profiles were closer to reaching equilibrium. It is shown in **Fig. 5** (D) that at t_1 the maximum temperature difference between inlet and outlet in the adsorption section was about 33°C, and the temperature decreased from t_1 to t_{10} because of the cooling effect of process air. It is shown in **Fig. 5** (E) that from inlet to outlet, the CO_2 concentration decreased. This is because the CO_2 in the gas phase was gradually

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captured by the adsorbent. From t_1 to t_3, the CO_2 concentration decreased because the adsorption capacity was large under the decreasing temperature. From t_3 to t_{10}, the CO_2 concentration increased because the adsorbent gradually reached saturation, which is also shown in **Fig. 5** (F). Besides, it is shown in **Fig. 5** (F) that from inlet to outlet, the CO_2 uptake decreased because the CO_2 concentration on the back end of channel was lower than that on the front side.

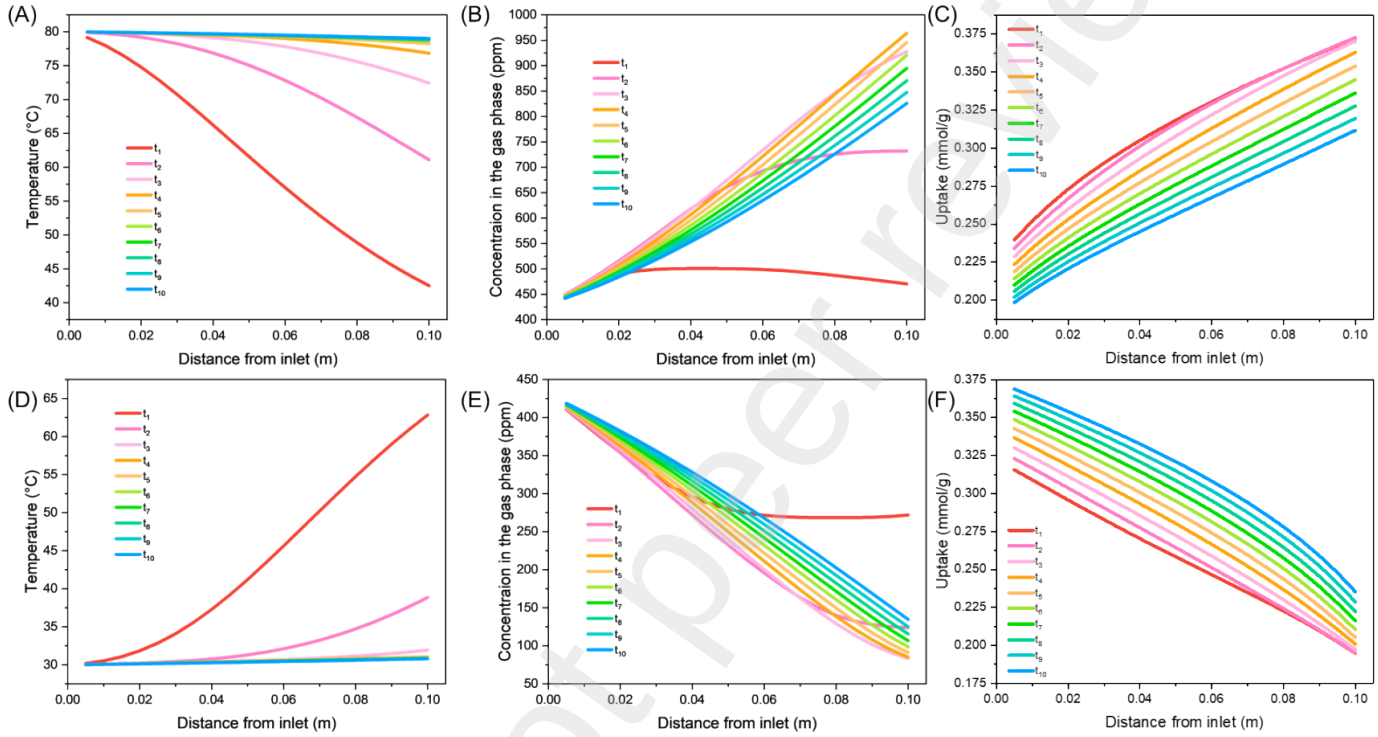


Fig. 5 The parameter profiles in the channel

(A) The temperature variation from the channel inlet to outlet in the regeneration section.

(B) The CO_2 concentration variation from the channel inlet to outlet in the regeneration section.

(C) The CO_2 uptake variation from the channel inlet to outlet in the regeneration section.

(D) The temperature variation from the channel inlet to outlet in the adsorption section.

(E) The CO_2 concentration variation from the channel inlet to outlet in the adsorption section.

(F) The CO_2 uptake variation from the channel inlet to outlet in the adsorption section.

There are many factors that can influence the performance of a rotating bed. Investigating each one through experiments can be time-consuming and, in some cases, challenging. For example, structural parameters like the regeneration section area ratio (*RSAR*) can be relatively difficult to adjust experimentally. Therefore, we used the model to perform parametric analysis. The effects of wheel thickness, *RSAR*, and channel size were investigated by the control variable method and elucidated in **Supplemental Information**. Herein we present a multiparameter optimization using the orthogonal test method.[75] Nine parameters (linear

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density of adsorbent l_{ad}, linear density of substrate l_m, process air temperature T_{pro}, process air velocity v_{pro}, regeneration air temperature T_{reg}, regeneration air velocity v_{reg}, wheel rotating speed ω, wheel thickness L_{wheel}, and regeneration section area ratio *RSAR*) were selected for investigation, and three levels were set for each parameter. Without considering the interaction between parameters, at least 27 groups of tests should be performed, which was denoted as $L_{27}(3^9)$. The carbon capture rate (*CCR*), product stream average concentration (*PSC*), productivity (*PR*), and energy consumption (*EC*) were used as performance indicators, and results are shown in **Table S6**. It is observed that group No. 2, No. 12, No. 21, and No. 11 exhibited the highest *CCR*, highest *PSC*, highest *PR*, and lowest *EC*, respectively. Through range analysis, it is found in **Table S7** that the factors that have the most significant impact on *CCR*, *PSC*, *PR*, and *EC*, are *RSAR*, regeneration temperature, regeneration temperature, and wheel thickness, respectively. Additionally, **Table S7** provides the conditions for achieving the optimal indicators. If a comprehensive evaluation index I_{comp} ($= \overline{CCR} + \overline{PSC} + \overline{PR} - \overline{EC}$, defined in the **Supplemental Information**) was used as an indicator, the parameters that gave the best overall performance is l_{ad}=0.0015, l_m=0.0005, T_{pro}=15°C, v_{pro}=3 m s^{-1}, T_{reg}=90°C, v_{reg}=2 m s^{-1}, ω=5 rph, L_{wheel}=0.2 m, and *RSAR*=1/2. Through the orthogonal test, the parameters having the greatest impact on the performance were determined and the multi-objective optimization was achieved.

Techno-economic analysis

Renewable production from direct air capture of CO_2 is a highly desired goal. The above results prove that the rotary adsorber can continuously remove CO_2 from air and generate CO_2-rich air that can be utilized in the greenhouse. Meanwhile, the rotary adsorber can be driven at low temperature (60–100°C), making it possible to use waste heat as a source of energy. Herein we present a study on the integration of DAC with low-carbon energy and its application in vegetable farming, and a preliminary techno-economic analysis is conducted on the proposed flowsheet shown in **Fig. 6** (A). The main unit processes in the system include a

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rotary adsorber, an incineration plant that supplies the heat for regeneration, and a greenhouse with tomato plants grown inside. A 1000 m^2 greenhouse that requires 18000 m^3 h^{-1} inlet air with ~1200 ppm CO_2 concentration was considered. Assume that the rotary adsorber has a productivity of 342 kg per day with the regeneration temperature of 80°C, a rotary adsorber with a wheel diameter of 2650 mm and a thickness of 1300 mm will fit this system. The air at the regeneration flow path outlet of the adsorber is cooled before entering the greenhouse using a gas-gas heat exchanger, which also allows for heat recovery. **Table S8** provides the fluid status at different points in the system as marked in **Fig. 6** (A). The system is modeled as continuous system where the DAC automatically cycles through all the necessary steps and supplies CO_2-rich air during the day. The financial assumptions and detailed calculations are provided in **Supplemental Information**. On this basis, the costs and benefits of carbon capture were calculated and shown in **Fig. 6** (B-D).

The capital costs of the integrated system are dominated by the equipment costs, especially for the adsorber (**Fig. 6** (B)). The annual cash flow (**Fig. 6** (C)) shows that the operation costs are dominated by adsorbent cost and electricity consumption. When compared to the scenario without CO_2 enrichment, the integration of the rotary adsorber results in an extra annual net profit of \$15350–24175 $year^{-1}$, which leads to a capital recovery period of 2.4–3.8 years assuming the system operation of 20 years. To account for uncertainties in the selling price of the products and the plant production rate increase,[61,76] a sensitivity analysis was performed on these two factors and results are presented in **Fig. 6** (D). It is seen that with the increase in tomato selling price and in plant yield, the net annual profit increased up to \$70000 $year^{-1}$. The break-even curve had the shape of an arc, which shows diminishing yield increase is required to break even as tomato price increases. Overall, the adsorber captures around 112 t CO_2 from air per year with a cost of \$127–206 t^{-1}. This highlights both the environmental and economic benefits of the system.

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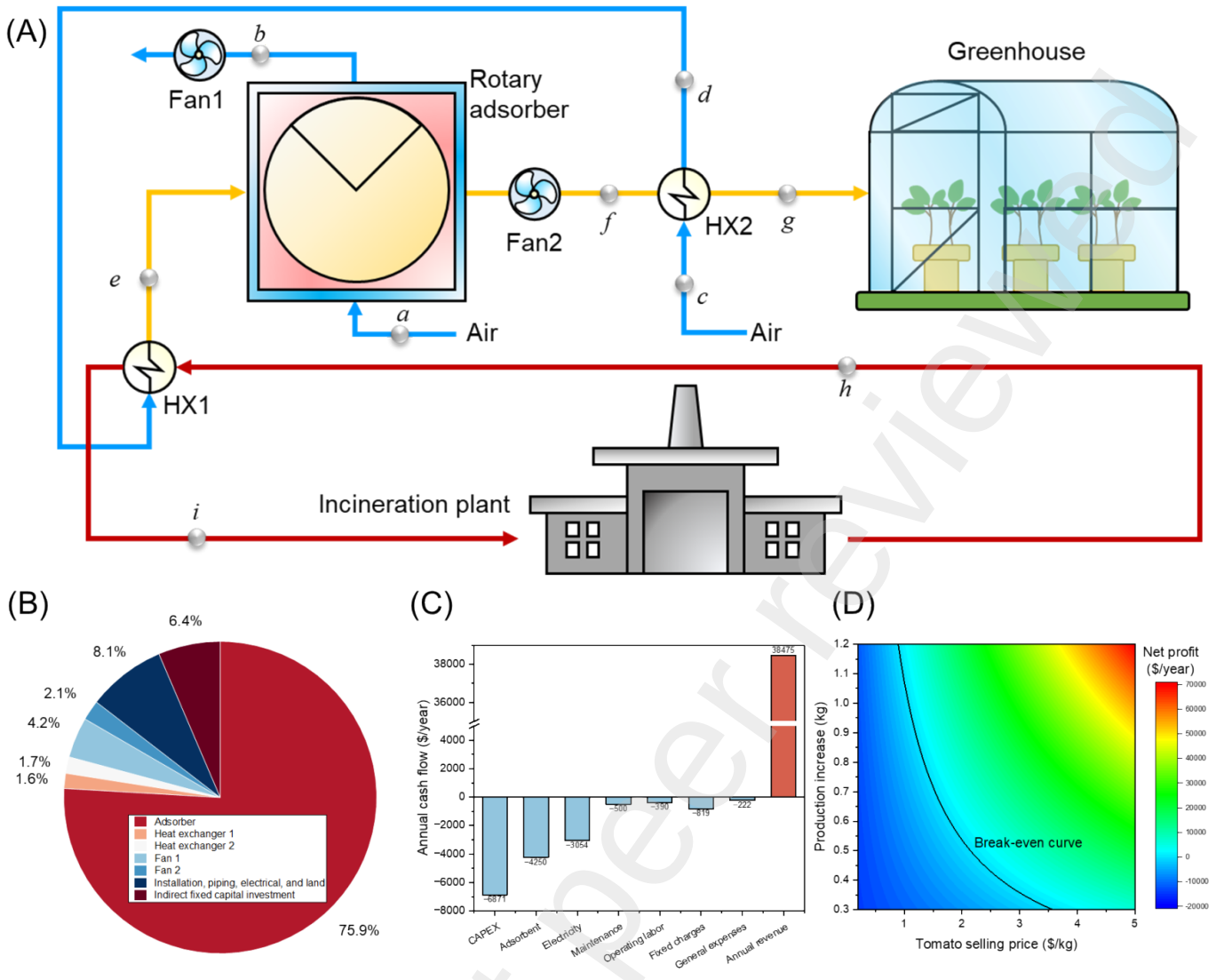


Fig. 6 Techno-economic analysis of the integrated system

(A) The schematic of greenhouse and rotary adsorber CO_2 enrichment system.

(B) The breakdown of capital costs of the CO_2 enrichment system.

(C) The annual cash flow of operating the CO_2 enrichment system.

(D) The net annual profit of the processes as a function of tomato selling price and tomato yield increase (the line represents the values for the system to break even).

Comparison with current DAC devices

The novel design of rotary adsorber endows it unique advantages compared to conventional carbon capture units. The rotary adsorber uses a low pressure drop structured adsorbent, so the energy consumption of fans can be significantly reduced. The enhanced heat and mass transfer performance in gas-solid contact results in a faster completion of the temperature swing adsorption cycle, taking only 8 minutes, and improves the capture efficiency. Based on the above experimental and simulation results, the comparison of the

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proposed rotary adsorber with traditional carbon capture devices in terms of key indicators are shown in **Fig. 7**. The high capture rate of the rotary adsorber is shown in **Fig. 7** (A). In terms of operational energy penalty, its consumption is relatively low, as shown in **Fig. 7** (B). The rotary adsorber demonstrates a fast productivity, as shown in **Fig. 7** (C-D), using both mass-based and volume-based metrics for comparison. Volumetric productivity is a relatively uncommon metric. It is difficult to directly find the values of each device in the literature, so the data in **Fig. 7** (D) are obtained indirectly by calculating the size and productivity of the adsorber. According to the study by Sabatino et al.[22], the volumetric productivity of adsorption-based DAC ranges from 0.13–10.6 kg $m^{-3}h^{-1}$. However, this is a simulated study and is partially based on hypothetical combinations of isotherms. The mass productivity of 0.664 g g^{-1} day^{-1} and volumetric productivity of 9.1 kg m^{-3} h^{-1} in this paper are the highest values among experimental research to the best of our knowledge. The rotary's ability to operate continuously within a single vessel reduces its footprint compared to traditional fixed beds and fluidized beds, making it a more feasible option for large-scale implementation. Furthermore, our study reports a relatively low capture cost (**Fig. 7** (E)), further supporting the economic feasibility of this device.

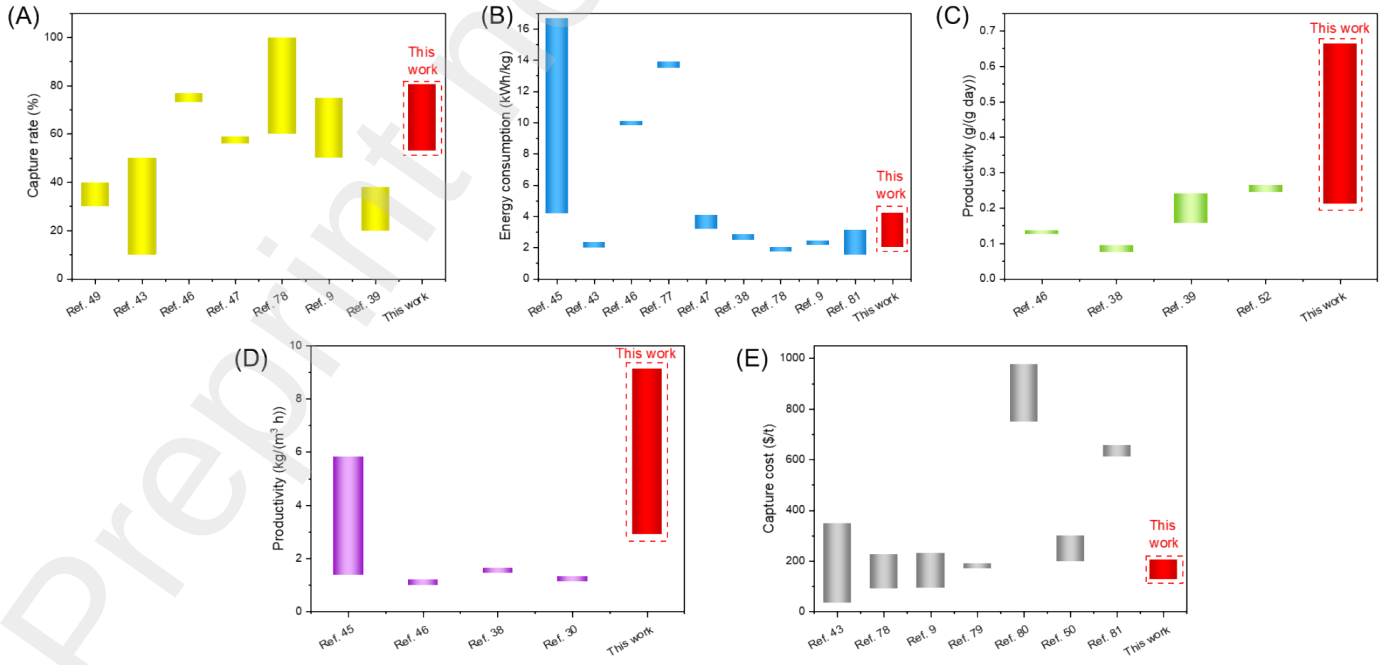


Fig. 7 The comparison of key evaluation indicators between the rotary adsorber and current DAC units[9,30,38,39,43,45-47,49,50,52,77-81]

(A) The CO_2 capture rate. These values were obtained by comparing the inlet and outlet CO_2 concentration of the capture device.

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(B) The CO_2 capture energy consumption. For comparison, the units were all directly converted to kWh kg^{-1}.
(C) The CO_2 productivity per unit mass. These values were obtained by dividing CO_2 production by time and adsorbent mass.
(D) The CO_2 productivity per unit volume. These values were obtained by dividing CO_2 production by time and air contactor volume.
(E) The CO_2 capture cost. The capture cost estimation in this study considers both the capital cost and operation cost.

DISCUSSION

In this study, we demonstrated a novel strategy of capturing CO_2 from air using the rotary adsorber. First, we developed structured adsorbents with desirable adsorption properties, low pressure drop, favorable heat and mass transfer, and low manufacture cost. Then, the laboratory-scale device was built to perform tests under ambient conditions. Experimental results show that the adsorber presented high carbon capture performance (50–80% capture rate in 50–100 m^3 h^{-1} 430 ppm CO_2 with 2–4.2 kWh kg^{-1} operation energy consumption). Next, we developed a mathematical model for the rotary adsorber under DAC conditions. On the basis of verifying the model's reliability with experimental data, we uncovered parameter distribution on the wheel outlet surface as well as inside the wheel channel. A parametric analysis was conducted using the control variable method to determine the impacts of individual parameters. Moreover, a comprehensive optimization of multiple parameters was carried out to identify the optimal conditions. Based on existing experimental and simulation data, the adsorber is combined with a greenhouse to form an integrated system in the techno-economic analysis. The adsorber utilizes waste heats to extract CO_2 from the air and deliver it to plants. The overall carbon capture cost and annual income were calculated, demonstrating positive environmental (\$127–206 $t^{-1}CO_2$ capture per year) and economic (extra net profit of \$15350–24175 per year) outcomes of the system.

According to key indicators for carbon capture devices, the approach proposed in this paper is compared with the existing research, and it is found that all the indicators in this study are desirable or superior, especially for the volumetric productivity. The advantages of rotary adsorber for DAC can be summarized as follows. 1) The implementation of structured adsorbents significantly reduces pressure drop, enabling high capture rates

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even at high air flow rates. The improved heat and mass transfer performance allows for low-temperature regeneration with quick cycles, resulting in efficient utilization of the adsorbent and low energy consumption. 2) An ingenious mechanical structure is designed to achieve continuous operation in a compact space, which eliminates the need for complex control strategies of the system and achieves ultra-high productivity (up to 0.664 g g^{-1} day^{-1} or 9.143 kg h^{-1} m^{-3}), thereby greatly reducing the device footprint and capture cost. 3) The device features a modular design, making it easily scalable with the use of structured adsorbents. It can also be manufactured as a mobile unit for deployment in areas with ample energy and/or where CO_2 storage and utilization are convenient. The combined effects of the above favorable factors lead to a relatively low capture cost \$127–206 t^{-1} of the adsorber.

As revealed by the model, the capture performance of the device can be further enhanced by increasing the adsorbent capacity and adjusting the device structure, and further testing of prototypes is required in the future. While this study focuses on the carbon capture performance of the rotary adsorber, further investigation of downstream applications is necessary to fully realize negative carbon effects. Integrated system experiments can be conducted to validate the design proposed in this paper. It is worth mentioning that though this article uses air purge for regeneration, it is theoretically feasible to expand its applications by adjusting the gas flow rate and regeneration gas type. One example is further reducing the regeneration air flow rate and/or increasing the return air amount to elevate the CO_2 concentration to 1–5% for microalgae cultivation or air-conditioned refrigerators. Alternatively, CO_2 with a purity greater than 90% can be obtained through steam purge and condensation for use in geological storage, fuel synthesis, or chemical material production. Given the flexible and efficient nature of rotary adsorber, it is conceivable that these systems will create more opportunities for sustainable development.

METHODS

Materials synthesis and characterizations

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Small scale (2 cm×2 cm×2 cm) structured adsorbents were synthesized using a method reported previously for characterizations.[82] The soaking solution was prepared by mixing liquid polyethyleneimine (PEI) and silica sol, and then it was used to coat the ceramic fiber honeycomb (Nichias, Japan). The structured adsorbents were obtained after the immersion, purging, and vacuum drying. The isotherms were acquired using the ASAP 2020 (Micromeritics) surface area and porosity analyzer. The adsorption and desorption curves of the samples were measured using the TGA 8000 (Perkin Elmer). In the TGA tests, 400 ppm CO_2 in N_2 was introduced into the sample chamber in the adsorption process and pure N_2 was used as the purge gas. The breakthrough curve of the adsorbents and the effect of water vapor on the adsorption performance were tested using a self-made fixed bed equipment, which is described in detail in the **Supplemental Information**.

Experiments of the device and data processing

A ceramic honeycomb (Φ400 mm×100 mm) was used as the substrate to obtain the wheel-shaped structured adsorbent. It was then installed in the rotary adsorber prototype that was constructed in the laboratory. The structural parameters of the prototype are show in **Table S3**. The tests of the prototype were performed during June and July in 2022 in the laboratory in Shanghai. The ambient air (30–37°C, 40–70%RH) was directly introduced into the prototype for adsorption and regeneration. Temperature and humidity sensors, anemometers, and CO_2 sensors were used to record the air status inside the flow paths. A variable frequency motor was used to control the rotation speed of the wheel. The CO_2 concentration and temperature data were used to calculate the key performance indicators, which is described in detail in the **Supplemental Information**.

Mathematical model development

The finite volume method was used to establish the DAC rotary adsorber model. Taking a single flow channel as the control volume, it was divided into the gas side and the solid side. It is assumed that all flow channels on the wheel were consistent, the substrate and adsorbent were evenly distributed, and the

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temperature and mass concentration gradients in the radial and circumferential directions of the gas flow can be neglected. Thus, the study was simplified into the solution of the one-dimensional heat and mass transfer model. The upwind scheme of backward difference was used to discretize the control equations, then the computer program was compiled to solve them. The model development details can be found in the **Supplemental Information.**

Techno-economic analysis

We estimate the cost of our proposed process by summing up the capital (CAPEX) and operating (OPEX) costs according to the procedure in chemical engineering economic texts[83]. The time zero CAPEX is calculated in terms of fixed capital investment, while annualized capital costs are evaluated considering a plant lifetime of 20 years with an interest rate of 10%. The OPEX is calculated as the sum of direct production costs, fixed charges, administrative costs, and distribution and marketing costs. The net annual profit is calculated by subtracting total costs from product sales. The financial assumptions and calculation details are provided in the **Supplemental Information**.

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Reference

1. Masson-Delmotte, V., Zhai, P., Pörtner, H.-O., Roberts, D., Skea, J., Shukla, P.R., Pirani, A., Moufouma-Okia, W., Péan, C., and Pidcock, R. (2019). Global warming of 1.5℃. IPCC.
2. Le Quéré, C., Andrew, R.M., Friedlingstein, P., Sitch, S., Pongratz, J., Manning, A.C., Korsbakken, J.I., Peters, G.P., Canadell, J.G., Jackson, R.B., et al. (2018). Global Carbon Budget 2017. Earth Syst. Sci. Data *10*, 405-448. 10.5194/essd-10-405-2018.
3. Bui, M., Adjiman, C.S., Bardow, A., Anthony, E.J., Boston, A., Brown, S., Fennell, P.S., Fuss, S., Galindo, A., Hackett, L.A., et al. (2018). Carbon capture and storage (CCS): the way forward. Energy & Environmental Science *11*, 1062-1176. 10.1039/c7ee02342a.
4. Smith, P., Davis, S.J., Creutzig, F., Fuss, S., Minx, J., Gabrielle, B., Kato, E., Jackson, R.B., Cowie, A., Kriegler, E., et al. (2015). Biophysical and economic limits to negative CO2 emissions. Nature Climate Change *6*, 42-50. 10.1038/nclimate2870.
5. Lindsey, R. (2022). Climate Change: Atmospheric Carbon Dioxide. https://www.climate.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide#:%7E:text=The%20global%20average%20atmospheric%20carbon,least%20the%20past%20800%2C000%20years.
6. Sanz-Perez, E.S., Murdock, C.R., Didas, S.A., and Jones, C.W. (2016). Direct Capture of CO2 from Ambient Air. Chem Rev *116*, 11840-11876. 10.1021/acs.chemrev.6b00173.
7. Creutzig, F., Breyer, C., Hilaire, J., Minx, J., Peters, G.P., and Socolow, R. (2019). The mutual dependence of negative emission technologies and energy systems. Energy & Environmental Science *12*, 1805-1817. 10.1039/C8EE03682A.
8. Fuss, S., Lamb, W.F., Callaghan, M.W., Hilaire, J., Creutzig, F., Amann, T., Beringer, T., de Oliveira Garcia, W., Hartmann, J., Khanna, T., et al. (2018). Negative emissions—Part 2: Costs, potentials and side effects. Environmental Research Letters *13*. 10.1088/1748-9326/aabf9f.
9. Keith, D.W., Holmes, G., St. Angelo, D., and Heidel, K. (2018). A Process for Capturing CO2 from the Atmosphere. Joule *2*, 1573-1594. 10.1016/j.joule.2018.05.006.
10. Kiani, A., Jiang, K., and Feron, P. (2020). Techno-Economic Assessment for CO2 Capture From Air Using a Conventional Liquid-Based Absorption Process. Frontiers in Energy Research *8*. 10.3389/fenrg.2020.00092.
11. Mazzotti, M., Baciocchi, R., Desmond, M.J., and Socolow, R.H. (2013). Direct air capture of CO2 with chemicals: optimization of a two-loop hydroxide carbonate system using a countercurrent air-liquid contactor. Climatic Change *118*, 119-135. 10.1007/s10584-012-0679-y.
12. Goeppert, A., Czaun, M., Surya Prakash, G.K., and Olah, G.A. (2012). Air as the renewable carbon source of the future: an overview of CO2 capture from the atmosphere. Energy & Environmental Science *5*. 10.1039/c2ee21586a.
13. Jones, C.W. (2011). CO(2) capture from dilute gases as a component of modern global carbon management. Annu Rev Chem Biomol Eng *2*, 31-52. 10.1146/annurev-chembioeng-061010-114252.
14. Darunte, L.A., Walton, K.S., Sholl, D.S., and Jones, C.W. (2016). CO2 capture via adsorption in amine-functionalized sorbents. Current Opinion in Chemical Engineering *12*, 82-90. 10.1016/j.coche.2016.03.002.
15. Fasihi, M., Efimova, O., and Breyer, C. (2019). Techno-economic assessment of CO2 direct air capture plants. Journal of Cleaner Production *224*, 957-980. 10.1016/j.jclepro.2019.03.086.
16. Aghel, B., Janati, S., Wongwises, S., and Shadloo, M.S. (2022). Review on CO2 capture by blended amine solutions. International Journal of Greenhouse Gas Control *119*. 10.1016/j.ijggc.2022.103715.
17. Mazari, S.A., Ghalib, L., Sattar, A., Bozdar, M.M., Qayoom, A., Ahmed, I., Muhammad, A., Abro,

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4388839

R., Abdulkareem, A., Nizamuddin, S., et al. (2020). Review of modelling and simulation strategies for evaluating corrosive behavior of aqueous amine systems for CO2 capture. International Journal of Greenhouse Gas Control *96*. 10.1016/j.ijggc.2020.103010.

18. Younas, M., Sohail, M., Leong, L.K., Bashir, M.J., and Sumathi, S. (2016). Feasibility of CO2 adsorption by solid adsorbents: a review on low-temperature systems. International Journal of Environmental Science and Technology *13*, 1839-1860. 10.1007/s13762-016-1008-1.
19. Madhu, K., Pauliuk, S., Dhathri, S., and Creutzig, F. (2021). Understanding environmental trade-offs and resource demand of direct air capture technologies through comparative life-cycle assessment. Nature Energy. 10.1038/s41560-021-00922-6.
20. Shi, X., Xiao, H., Azarabadi, H., Song, J., Wu, X., Chen, X., and Lackner, K.S. (2020). Sorbents for the Direct Capture of CO2 from Ambient Air. Angew Chem Int Ed Engl *59*, 6984-7006. 10.1002/anie.201906756.
21. Deutz, S., and Bardow, A. (2021). Life-cycle assessment of an industrial direct air capture process based on temperature–vacuum swing adsorption. Nature Energy *6*, 203-213. 10.1038/s41560-020-00771-9.
22. Sabatino, F., Grimm, A., Gallucci, F., van Sint Annaland, M., Kramer, G.J., and Gazzani, M. (2021). A comparative energy and costs assessment and optimization for direct air capture technologies. Joule *5*, 1-30. 10.1016/j.joule.2021.05.023.
23. Realmonte, G., Drouet, L., Gambhir, A., Glynn, J., Hawkes, A., Koberle, A.C., and Tavoni, M. (2019). An inter-model assessment of the role of direct air capture in deep mitigation pathways. Nat Commun *10*, 3277. 10.1038/s41467-019-10842-5.
24. Zhu, X., Xie, W., Wu, J., Miao, Y., Xiang, C., Chen, C., Ge, B., Gan, Z., Yang, F., Zhang, M., et al. (2022). Recent advances in direct air capture by adsorption. Chem Soc Rev. 10.1039/d1cs00970b.
25. Erans, M., Sanz-Pérez, E.S., Hanak, D.P., Clulow, Z., Reiner, D.M., and Mutch, G.A. (2022). Direct air capture: process technology, techno-economic and socio-political challenges. Energy & Environmental Science. 10.1039/d1ee03523a.
26. Miao, Y., He, Z., Zhu, X., Izikowitz, D., and Li, J. (2021). Operating temperatures affect direct air capture of CO2 in polyamine-loaded mesoporous silica. Chemical Engineering Journal *426*. 10.1016/j.cej.2021.131875.
27. Kumar, D.R., Rosu, C., Sujan, A.R., Sakwa-Novak, M.A., Ping, E.W., and Jones, C.W. (2020). Alkyl-Aryl Amine-Rich Molecules for CO2 Removal via Direct Air Capture. ACS Sustainable Chemistry & Engineering. 10.1021/acssuschemeng.0c03706.
28. Kwon, H.T., Sakwa-Novak, M.A., Pang, S.H., Sujan, A.R., Ping, E.W., and Jones, C.W. (2019). Aminopolymer-Impregnated Hierarchical Silica Structures: Unexpected Equivalent CO2 Uptake under Simulated Air Capture and Flue Gas Capture Conditions. Chemistry of Materials *31*, 5229-5237. 10.1021/acs.chemmater.9b01474.
29. Zhu, X., Lyu, M., Ge, T., Wu, J., Chen, C., Yang, F., O'Hare, D., and Wang, R. (2021). Modified layered double hydroxides for efficient and reversible carbon dioxide capture from air. Cell Reports Physical Science *2*, 100484. https://doi.org/10.1016/j.xcrp.2021.100484.
30. Zhu, X., Ge, T., Yang, F., Lyu, M., Chen, C., O'Hare, D., and Wang, R. (2020). Efficient CO2 capture from ambient air with amine-functionalized Mg–Al mixed metal oxides. Journal of Materials Chemistry A *8*, 16421-16428. 10.1039/d0ta05079b.
31. Zhao, M., Xiao, J., Gao, W., and Wang, Q. (2022). Defect-rich Mg-Al MMOs supported TEPA with enhanced charge transfer for highly efficient and stable direct air capture. Journal of Energy Chemistry *68*, 401-410. 10.1016/j.jechem.2021.12.031.
32. McDonald, T.M., Lee, W.R., Mason, J.A., Wiers, B.M., Hong, C.S., and Long, J.R. (2012). Capture of carbon dioxide from air and flue gas in the alkylamine-appended metal-organic framework mmen-

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4388839

Mg2(dobpdc). J Am Chem Soc *134*, 7056-7065. 10.1021/ja300034j.
33. Rim, G., Kong, F., Song, M., Rosu, C., Priyadarshini, P., Lively, R.P., and Jones, C.W. (2022). Sub-Ambient Temperature Direct Air Capture of CO2 using Amine-Impregnated MIL-101(Cr) Enables Ambient Temperature CO2 Recovery. JACS Au *2*, 380-393. 10.1021/jacsau.1c00414.
34. Wu, J., Zhu, X., Yang, F., Wang, R., and Ge, T. (2022). Shaping techniques of adsorbents and their applications in gas separation: a review. Journal of Materials Chemistry A *10*, 22853-22895. 10.1039/d2ta04352a.
35. Sakwa-Novak, M.A., Yoo, C.J., Tan, S., Rashidi, F., and Jones, C.W. (2016). Poly(ethylenimine)-Functionalized Monolithic Alumina Honeycomb Adsorbents for CO2 Capture from Air. ChemSusChem *9*, 1859-1868. 10.1002/cssc.201600404.
36. Lee, Y.H., Kwon, Y., Kim, C., Hwang, Y.E., Choi, M., Park, Y., Jamal, A., and Koh, D.Y. (2021). Controlled Synthesis of Metal-Organic Frameworks in Scalable Open-Porous Contactor for Maximizing Carbon Capture Efficiency. JACS Au *1*, 1198-1207. 10.1021/jacsau.1c00068.
37. Thakkar, H., Eastman, S., Hajari, A., Rownaghi, A.A., Knox, J.C., and Rezaei, F. (2016). 3D-Printed Zeolite Monoliths for CO2 Removal from Enclosed Environments. ACS Appl Mater Interfaces *8*, 27753-27761. 10.1021/acsami.6b09647.
38. Wurzbacher, J.A., Gebald, C., and Steinfeld, A. (2011). Separation of CO2 from air by temperature-vacuum swing adsorption using diamine-functionalized silica gel. Energy & Environmental Science *4*. 10.1039/c1ee01681d.
39. Zhu, X., Ge, T., Yang, F., and Wang, R. (2021). Design of steam-assisted temperature vacuum-swing adsorption processes for efficient CO2 capture from ambient air. Renewable and Sustainable Energy Reviews *137*. 10.1016/j.rser.2020.110651.
40. Shi, X., Xiao, H., Kanamori, K., Yonezu, A., Lackner, K.S., and Chen, X. (2020). Moisture-Driven CO2 Sorbents. Joule *4*, 1823-1837. 10.1016/j.joule.2020.07.005.
41. Diederichsen, K.M., Liu, Y., Ozbek, N., Seo, H., and Hatton, T.A. (2022). Toward solvent-free continuous-flow electrochemically mediated carbon capture with high-concentration liquid quinone chemistry. Joule *6*, 221-239. 10.1016/j.joule.2021.12.001.
42. Voskian, S., and Hatton, T.A. (2019). Faradaic electro-swing reactive adsorption for CO2 capture. Energy & Environmental Science *12*, 3530-3547. 10.1039/c9ee02412c.
43. Sadiq, M.M., Batten, M.P., Mulet, X., Freeman, C., Konstas, K., Mardel, J.I., Tanner, J., Ng, D., Wang, X., Howard, S., et al. (2020). A Pilot-Scale Demonstration of Mobile Direct Air Capture Using Metal-Organic Frameworks. Advanced Sustainable Systems *4*. 10.1002/adsu.202000101.
44. Eric Ping, M.S.-N., Peter Eisenberger (2018). Global Thermostat Low Cost Direct Air Capture Technology. International Conference on Negative CO2 Emissions.
45. Schellevis, H.M., van Schagen, T.N., and Brilman, D.W.F. (2021). Process optimization of a fixed bed reactor system for direct air capture. International Journal of Greenhouse Gas Control *110*, 103431. https://doi.org/10.1016/j.ijggc.2021.103431.
46. E. Bajamundi, C.J., Koponen, J., Ruuskanen, V., Elfving, J., Kosonen, A., Kauppinen, J., and Ahola, J. (2019). Capturing CO2 from air: Technical performance and process control improvement. Journal of CO2 Utilization *30*, 232-239. 10.1016/j.jcou.2019.02.002.
47. Wurzbacher, J.A., Gebald, C., Piatkowski, N., and Steinfeld, A. (2012). Concurrent separation of CO2 and H2O from air by a temperature-vacuum swing adsorption/desorption cycle. Environ Sci Technol *46*, 9191-9198. 10.1021/es301953k.
48. Christoph Gebald, W.M., Nicolas Repond, Tobias Ruesch, and Jan André Wurzbacher (2019) Direct air capture device. Mar . 19, 2019.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4388839

49. Yu, Q., and Brilman, W. (2020). A Radial Flow Contactor for Ambient Air CO2 Capture. Applied Sciences *10*. 10.3390/app10031080.
50. McQueen, N., Psarras, P., Pilorge, H., Liguori, S., He, J., Yuan, M., Woodall, C.M., Kian, K., Pierpoint, L., Jurewicz, J., et al. (2020). Cost Analysis of Direct Air Capture and Sequestration Coupled to Low-Carbon Thermal Energy in the United States. Environ Sci Technol *54*, 7542-7551. 10.1021/acs.est.0c00476.
51. House, K.Z., Baclig, A.C., Ranjan, M., van Nierop, E.A., Wilcox, J., and Herzog, H.J. (2011). Economic and energetic analysis of capturing CO2 from ambient air. Proc Natl Acad Sci U S A *108*, 20428-20433. 10.1073/pnas.1012253108.
52. Gebald, C., Wurzbacher, J.A., Tingaut, P., Zimmermann, T., and Steinfeld, A. (2011). Amine-based nanofibrillated cellulose as adsorbent for CO2 capture from air. Environ Sci Technol *45*, 9101-9108. 10.1021/es202223p.
53. Joel, A.S., Wang, M., Ramshaw, C., and Oko, E. (2017). Modelling, simulation and analysis of intensified regenerator for solvent based carbon capture using rotating packed bed technology. Applied Energy *203*, 11-25. 10.1016/j.apenergy.2017.05.157.
54. Yang, H., Fang, X., Li, Z., Sun, H., and Chen, H. (2022). Copper-Doped Small Pore Zeolites for CO2 Capture by Honeycomb Rotor with Low Temperature Regeneration. ACS Sustainable Chemistry & Engineering *10*, 1759-1764. 10.1021/acssuschemeng.1c08347.
55. Herraiz, L., Palfi, E., Sánchez Fernández, E., and Lucquiaud, M. (2020). Rotary Adsorption: Selective Recycling of CO2 in Combined Cycle Gas Turbine Power Plants. Frontiers in Energy Research *8*. 10.3389/fenrg.2020.482708.
56. Ge, T.S., Dai, Y.J., and Wang, R.Z. (2014). Review on solar powered rotary desiccant wheel cooling system. Renewable and Sustainable Energy Reviews *39*, 476-497. 10.1016/j.rser.2014.07.121.
57. Sheng, Y., Zhang, L., Wang, Y., and Fang, L. (2019). Explore energy saving operation strategy: Indoor VOCs removal performance of silica gel rotor in clean-air heat pump system at low regeneration air temperature. Energy and Buildings *202*. 10.1016/j.enbuild.2019.109379.
58. Inc, S.T. (2023). Meet the Svante Carbon Capture Ecosystem. https://svanteinc.com/carbon-capture-technology/.
59. Nations, U. (2019). World Population Prospects 2019: Highlights.
60. Costantino, A., Comba, L., Sicardi, G., Bariani, M., and Fabrizio, E. (2021). Energy performance and climate control in mechanically ventilated greenhouses: A dynamic modelling-based assessment and investigation. Applied Energy *288*. 10.1016/j.apenergy.2021.116583.
61. M.Mortensen, L. (1987). Review: CO2 enrichment in greenhouses. Crop responses. Scientia Horticulturae *33*, 1-25.
62. Hao, P.-F., Qiu, C.-W., Ding, G., Vincze, E., Zhang, G., Zhang, Y., and Wu, F. (2020). Agriculture organic wastes fermentation CO2 enrichment in greenhouse and the fermentation residues improve growth, yield and fruit quality in tomato. Journal of Cleaner Production *275*. 10.1016/j.jclepro.2020.123885.
63. Poudel, M., and Dunn, B. (2017). Greenhouse carbon dioxide supplementation. Oklahoma Cooperative Extension Service.
64. Zhang, Y., Yasutake, D., Hidaka, K., Okayasu, T., Kitano, M., and Hirota, T. (2022). Crop-localised CO2 enrichment improves the microclimate, photosynthetic distribution and energy utilisation efficiency in a greenhouse. Journal of Cleaner Production *371*. 10.1016/j.jclepro.2022.133465.
65. Zhang, Y., Yasutake, D., Hidaka, K., Kitano, M., and Okayasu, T. (2020). CFD analysis for evaluating and optimizing spatial distribution of CO2 concentration in a strawberry greenhouse under different CO2 enrichment methods. Computers and Electronics in Agriculture *179*. 10.1016/j.compag.2020.105811.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4388839

66. Serna-Guerrero, R., Belmabkhout, Y., and Sayari, A. (2010). Modeling CO2 adsorption on amine-functionalized mesoporous silica: 1. A semi-empirical equilibrium model. Chemical Engineering Journal *161*, 173-181. 10.1016/j.cej.2010.04.024.
67. Chaikittisilp, W., Khunsupat, R., Chen, T.T., and Jones, C.W. (2011). Poly(allylamine)–Mesoporous Silica Composite Materials for CO2 Capture from Simulated Flue Gas or Ambient Air. Industrial & Engineering Chemistry Research *50*, 14203-14210. 10.1021/ie201584t.
68. Wu, J., Zhu, X., Chen, Y., Wang, R., and Ge, T. (2022). The analysis and evaluation of direct air capture adsorbents on the material characterization level. Chemical Engineering Journal *450*. 10.1016/j.cej.2022.137958.
69. Hou, P.-X., Orikasa, H., Yamazaki, T., Matsuoka, K., Tomita, A., Setoyama, N., Fukushima, Y., and Kyotani, T. (2005). Synthesis of Nitrogen-Containing Microporous Carbon with a Highly Ordered Structure and Effect of Nitrogen Doping on H2O Adsorption. Chemistry of Materials *17*, 5187-5193. 10.1021/cm051094k.
70. Kolle, J.M., Fayaz, M., and Sayari, A. (2021). Understanding the Effect of Water on CO2 Adsorption. Chem Rev *121*, 7280-7345. 10.1021/acs.chemrev.0c00762.
71. Zhou, X., and Reece, R. (2019). Experimental investigation for a non-adiabatic desiccant wheel with a concentric structure at low regeneration temperatures. Energy Conversion and Management *201*. 10.1016/j.enconman.2019.112165.
72. Fong, K.F., and Lee, C.K. (2018). Impact of adsorbent characteristics on performance of solid desiccant wheel. Energy *144*, 1003-1012. 10.1016/j.energy.2017.12.113.
73. Prasad, A.K., Yadav, L., and Yadav, A. (2019). Comparative analysis of different design of rotary dehumidifier. Heat Transfer-Asian Research *48*, 2193-2215. 10.1002/htj.21480.
74. Yamaguchi, S., and Saito, K. (2013). Numerical and experimental performance analysis of rotary desiccant wheels. International Journal of Heat and Mass Transfer *60*, 51-60. 10.1016/j.ijheatmasstransfer.2012.12.036.
75. Zheng, X., Yao, H., and Huang, Y. (2017). Orthogonal numerical simulation on multi-factor design for rapid pressure swing adsorption. Adsorption *23*, 685-697. 10.1007/s10450-017-9886-1.
76. Nilsen, S., Hovland, K., Dons, C., and Sletten, S.P. (1983). Effect of CO2 enrichment on photosynthesis, growth and yield of tomato. Scientia Horticulturae *20*, 1-14.
77. Santori, G., Charalambous, C., Ferrari, M.-C., and Brandani, S. (2018). Adsorption artificial tree for atmospheric carbon dioxide capture, purification and compression. Energy *162*, 1158-1168. 10.1016/j.energy.2018.08.090.
78. Zhang, W., Liu, H., Sun, C., Drage, T.C., and Snape, C.E. (2014). Capturing CO2 from ambient air using a polyethyleneimine–silica adsorbent in fluidized beds. Chemical Engineering Science *116*, 306-316. 10.1016/j.ces.2014.05.018.
79. Datta, A., and Krishnamoorti, R. (2023). Analysis of Direct Air Capture Integrated with Wind Energy and Enhanced Oil Recovery. Environ Sci Technol. 10.1021/acs.est.2c05194.
80. Leonzio, G., Fennell, P.S., and Shah, N. (2022). A Comparative Study of Different Sorbents in the Context of Direct Air Capture (DAC): Evaluation of Key Performance Indicators and Comparisons. Applied Sciences *12*. 10.3390/app12052618.
81. Wijesiri, R.P., Knowles, G.P., Yeasmin, H., Hoadley, A.F.A., and Chaffee, A.L. (2019). Technoeconomic Evaluation of a Process Capturing CO2 Directly from Air. Processes *7*. 10.3390/pr7080503.
82. Wu, J., Zhu, X., Yang, F., Ge, T., and Wang, R. (2021). Easily-synthesized and low-cost amine-functionalized silica sol-coated structured adsorbents for CO2 capture. Chemical Engineering Journal *425*. 10.1016/j.cej.2021.131409.
83. Peter, M.S.T., K.D (1991). Plant design and economics for chemical engineers.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4388839

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Supplemental Information

A rotary direct air capture adsorber with ultrahigh productivity for efficient CO_2 enrichment in greenhouse

Junye Wu[1, 2, 3], Kuihua Wang[1], Junde Zhao[1], Yanlin Chen[1], Zhuozhen Gan[1], Xuancan Zhu[1], Ruzhu Wang[1], Bo Wang[2, 3], Chi-Hwa Wang[2, 4], Yen Wah Tong[2, 4], and Tianshu Ge[1, *]

[1] Research Center of Solar Power & Refrigeration, Institute of Refrigeration and Cryogenics, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai, 200240, China
[2] Energy and Environmental Sustainability Solutions for Megacities (E2S2), Campus for Research Excellence and Technological Enterprise (CREATE), 138602, Singapore
[3] NUS Environmental Research Institute, National University of Singapore, Singapore 138602, Singapore
[4] Department of Chemical and Biomolecular Engineering, National University of Singapore, 4 Engineering Drive 4, Singapore 117585, Singapore

[*] Correspondence: baby_wo@sjtu.edu.cn (T. G.)

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Literature review

The studies about adsorption-based direct air capture devices are shown in **Table S1**. The studies about rotary beds are shown in **Table S2**.

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Table S1 The studies about adsorption-based direct air capture devices.

Cycle	Adsorbent	Adsorber	Adsorption condition	Desorption condition	Capture rate	Purity	Productivity	Energy consumption	Remark	Ref.
TVSA	Diamine-grafted silica gel	Fixed bed	400–440 ppm CO_2, 10–30°C, 0–80% RH, 2.5 L min^{-1}	10–150 mbar vacuum, 74–90°C	–	97.6%	158 mL of CO_2 (6.8 mL g^{-1} sorbent) per cycle (255 min)	439.6 kJ mol^{-1}	(Experiments) operated stably for 40 cycles	[1]
TVSA	Amine-functionalized nanofibrilated cellulose	Fixed bed	410–500 ppm CO_2, 10–30°C, 20–80% RH, 4 L min^{-1}	50 mbar, vacuum, 95°C	⩽58%	94.4%	–	505.5–652.5 kJ mol^{-1}	(Experiments) also measured the working capacity of H_2O; operated stably for 10 cycles	[2]
TVSA	Amine-functionalized polystyrene resin	Fixed bed	~470 ppm CO_2, 10–30°C, 30–80% RH, <760 m^3 h^{-1}	<0.15 bar vacuum, 60–80°C	~75%	> 95%	3.8 kg d^{-1}	10 kW h kg^{-1}	(Experiments) also produced water at molar ratio of 3.9 mol H_2O per mol of CO_2	[3]
TVSA	Proprietary MOF	Fixed bed	~400 ppm CO_2, 25°C, 50 m^3 h^{-1}	37 mbar, vacuum	10–50%	70–80%	0.143 kg d^{-1}	2.28 kW h kg^{-1}	(Experiments) operational cost \$35–350 t^{-1}, operated stably during 2680 cycles	[4]
TVSA	Lewatit® VP OC 1065	Fixed bed	~400 ppm CO_2, ~20°C, 70% RH, 0.08–0.15 m s^{-1}	350 mbar vacuum, 90°C 5–50 L min^{-1} N_2	–	<15%	0.1–0.42 kg d^{-1}	15–60 MJ kg^{-1}	(Simulations and experiments) examined the influence of weather conditions on the adsorption rate	[5]
TSA	Amine-modified silica	Fluidized bed	393.5 ppm CO_2, 20°C, 0.9–1.4% H_2O, 8 L min^{-1}	130°C, 8 L min^{-1} N_2	>60%	<6.5%	5.61–9.30 g day^{-1}	6.6 GJ t^{-1}	(Experiments) operational cost of \$91–227 t^{-1}	[6]
TSA	Lewatit® VP OC 1065	Moving bed	436 ppm, 19–22°C, 40–65% RH, 188 m^3 h^{-1}	120°C, 3 m^3 h^{-1} N_2	30–40%	–	–	0.7–1.5 GJ t^{-1} (contacting energy)	(Experiments) the device was operated as intermittently and continuously moving bed	[7]
TSA	Mg-MOF-74/13X/activated carbon	Rotary bed	400 ppm, CO_2, –20°C, 0.1–0.6 m s^{-1}	400 ppm, CO_2, 50–170°C, 0.1 m s^{-1}	–	430–550 ppm	–	–	(Simulations) direct air capture for CO_2 enrichment in greenhouse	[8]

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Table S2 The studies about rotary beds.

Sorbent	Description	Process condition	Regeneration condition	Capture rate	Purity	Productivity	Energy consumption	Remark	Ref.
Monoethanolamine	Absorption, CO_2 capture	3.5–4.5% CO_2, 10.8 mL min^{-1}	–	83–98%	–	–	–	(Experiments) showed the advantages of using rotating packed beds in terms of saving size and space and efficient operation	[9]
Monoethanolamine	Absorption, CO_2 capture	3–5% CO_2, 2.97 kmol h^{-1}, 50°C	–	83–98.2%	–	–	–	(Simulations) absorber volume reduction factor of 12 times was found	[10]
Monoethanolamine	Absorption, CO_2 capture	3.5–4.5% CO_2, 10.8 mL min^{-1}	Reboiler temperature 105–120°C	–	–	–	4–7 GJ t^{-1}	(Simulations) absorber showed a size reduction of 9.691 times	[11]
Metal oxide	Chemical looping combustion, CO_2 capture	50–200 mL min^{-1}, CH_4, 700°C	200–800 mL min^{-1} Air, 700°C	50%	85%	–	–	(Simulations and experiments) CuO/Al_2O_3 based oxygen carrier was selected, radial gas flow	[12]
Metal oxide	Chemical looping combustion, CO_2 capture	12.5-58% CH_4, 450–650°C, 10 atm	Air, 450–650°C, 10 atm	>95%	–	–	–	(Simulations) >99% fuel conversion	[13, 14]
Silica gel-haloid	Adsorption, dehumidification	14.3 g H_2O kg^{-1} air, 35°C, 360 m^3 h^{-1}	Air, 50–90°C, 135 m^3 h^{-1}	–	–	–	–	(Simulations and experiments) Low regeneration temperature, high COP_{th} (0.68–1.39), reduced size	[15]
High silica zeolite	Adsorption, VOCs capture	50–700 ppm acetone, <80% RH, 2–4 m s^{-1}, 25–35°C	Air, 180°C, 1–4 m s^{-1}	96–99%	–	500–7000 ppm acetone	–	(Experiments) self-sufficient system without any external heat source	[16]
Silica gel	Adsorption, VOCs capture	4.6–10.6 g kg^{-1} formaldehyde, 24°C, 936 m^3 h^{-1}	Air, 40–60°C, 432 m^3 h^{-1}	20–65%	–	–	Heat pump power 1.04–2.15 kW, COP 2.28–3.26	(Experiments) air cleaning efficiency decreased by 5–15% when regeneration air temperature decreased by 10°C	[17]
Activated carbon/13X	Adsorption, CO_2 capture	10.6% CO_2, 70°C, 30 L min^{-1}	Steam, 270°C, 12 bar	60.3–94.7%	93–95.6%	–	85.7 kcal kWh^{-1}	(Simulations) the auxiliary power usage of a 500 MW thermal power unit will be increased by about 0.095% with the addition of RBA, CO_2 avoided cost \$8.48 t^{-1}	[18]
13X	Adsorption, CO_2 capture	72.88 L min^{-1} kg^{-1}, 5% CO_2, 35°C	51.69 L min^{-1} kg^{-1}, steam, 125°C	90%	85.5%	–	6.45 kg steam per kg CO_2 captured	(Simulations) with a 10°C N_2 purge cooling section	[19]
PEI-MCM-41	Adsorption, CO_2 capture	42.38 L min^{-1} kg^{-1}, 5% CO_2, 35°C	51.69 L min^{-1} kg^{-1}, steam, 125°C	90%	78.3%	–	11.1 kg steam per kg CO_2 captured	(Simulations) with a 10°C N_2 purge cooling section	
CALF-20	Adsorption, CO_2	17% CO_2, 50°C	Steam purge, 110°C	70–80%	95%	10.2 t day^{-1}	1.9 kg kg^{-1} steam	(Experiments) < 1.5 min cycle time	[20]

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	capture						m^{-3}			
Activated carbon/13X	Adsorption, capture	CO_2	8–14% CO_2, 45°C, 200–659 kg s^{-1}	Air, 16.9°C, 560–585 kg s^{-1}	90%	–	–	–	(Simulations) selective exhaust gas recirculation, wheels of 24 m diameter and 2 m length are required	[21]
Activated carbon	Adsorption, capture	CO_2	800 kg s^{-1}, 20% CO_2, 30°C	320 kg s^{-1} steam, 200°C	<85%	<20%	–	–	(Simulations) with a cooling sector and a rinsing sector	[22]
Amine-containing nanogel particles	Adsorption, capture	CO_2	12% CO_2, 30°C, 0.18 m^3 h^{-1}	Vacuum, 75°C	93.4%	99.5%	0.622 kg kg $^{-1}$ h^{-1}	567 kJ kg^{-1}	(Simulations) with cycling of concentrated CO_2 to increase the CO_2 purity	[23]
Mg-MOF-74/13X/activated carbon	Adsorption, capture	CO_2	400 ppm, CO_2, –20°C, 0.1–0.6 m s^{-1}	400 ppm, CO_2, 50–170°C, 0.1 m s^{-1}	–	430–450 ppm	–	–	(Simulations) direct air capture for CO_2 enrichment in greenhouse	[8]
Copper-doped zeolite	Adsorption, capture	CO_2	80% CO_2, 25°C	65°C heat source	–	–	–	–	(Simulations) conceptual study, dehumidification before the CO_2 capture, circling gas was used	[24]

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Experimental section

Isotherm fitting

The fitted adsorption isotherms are shown in **Fig. S1**. The isotherm at 25°C was fitted by dual-site Toth model[25], and isotherms at 45–95°C were fitted by dual-site Langmuir-Freundlich model[26].

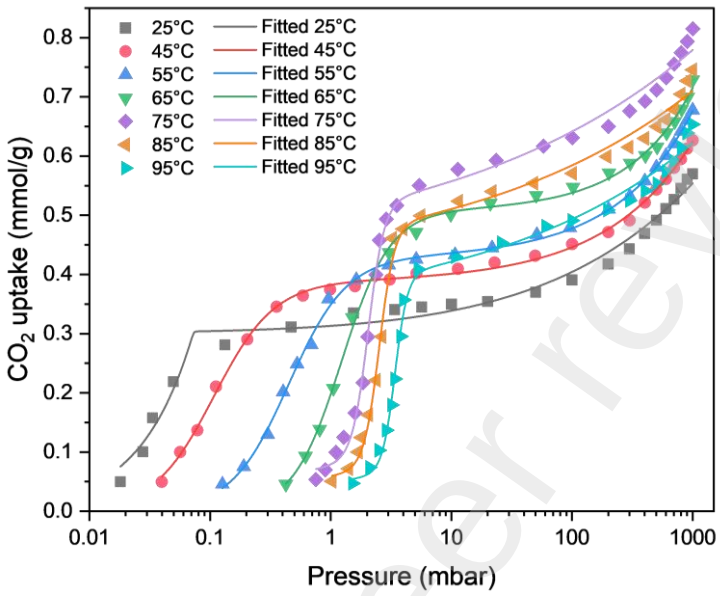


Fig. S1 The fitted isotherms at different temperatures.

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Breakthrough curve test

The self-made fixed bed described in previous work [27] with a height of 120 mm and internal diameter of 6 mm was used to evaluate the adsorption performance of the sample in simulated air in the absence/presence of moisture. The scrubbing bottle in a thermostatic water bath was used to generate moisture for the inlet gases, where the water content before the fixed bed was monitored using temperature and humidity sensors. The fixed bed as a stainless-steel tube was filled 0.1 g sample. The CO_2 concentration in the outlet gas after passing through a $CaCl_2$ drying tube was continuously detected by an THA100S non-dispersive infrared analyzer (0−600 ppm, ±2% full scale).

During the test, the fixed bed was first heated by a temperature control stove to 100°C followed by a counter-current N_2 purge with a flow rate of 100 mL min^{-1} for 60 min. The circulating oil bath was then opened to rapidly cool down the fixed bed to 25 °C. In the adsorption step, 400 ppm CO_2 in N_2 with a flow rate of 100 mL min^{-1} was co-currently added into the fixed bed for 120–400 min until the adsorption reached equilibrium. The CO_2 adsorption capacity q (mmol/g) in the tests was calculated according to **Equation S1**.

$$q = \frac{\int_0^{t_{ads}} (C_{in} - C_{out})\dot{Q}dt_{ads}}{m_{sample}} \quad \text{(S1)}$$

where C_{in} and C_{out} is the CO_2 concentration (ppm) in inlet and outlet gas respectively, $\dot{Q}$ is the gas flow rate (mL min^{-1}), t_{ads} is the adsorption time (min), m_{sample} is the sample mass (g).

The adsorption breakthrough curves of structured adsorbents under different relative humidity (RH) are shown in **Fig. S2**.

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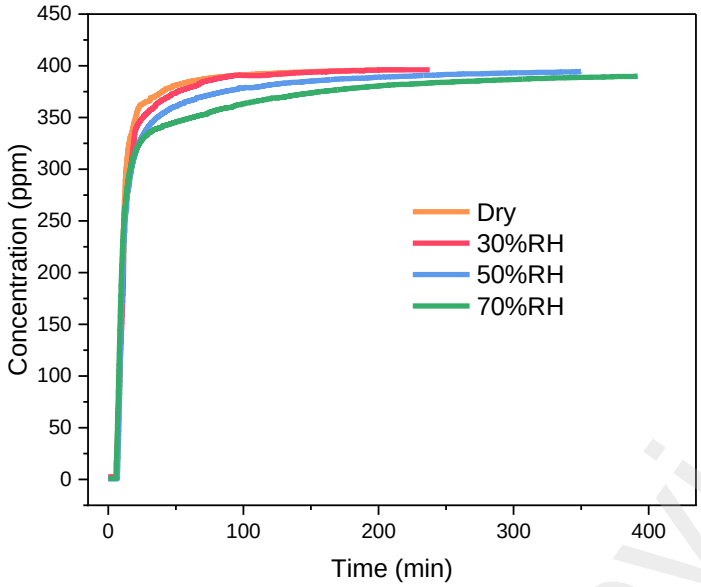


Fig. S2 The adsorption breakthrough curves of structured adsorbents at 25°C under different relative humidity.

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Prototype structural parameters

Structural parameters of the rotary adsorber prototype are shown in **Table S3**.

Table S3 Structural and operation parameters of rotary adsorber prototype.

Parameter	Value
Wheel diameter	400 mm
Wheel thickness	100 mm
Area ratio of adsorption section	1/3
Area ratio of regeneration section	2/3
Process air flow rate	50–100 $m^3 h^{-1}$
Process air temperature	30–40°C
Regeneration air flow rate	83 $m^3 h^{-1}$
Regeneration temperature	60–100°C
Return air flow rate	47 $m^3 h^{-1}$
Wheel rotation speed	2–7 rph
Channel height	0.0010 m
Channel width	0.0015 m
Structured adsorbent mass	4154 g

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Unsteady status operation

Once the device was activated, the motor drove the wheel to rotate, fans drew ambient air into flow paths, the heater heated the regeneration air, and the adsorbent started to capture and release CO_2. At first, the device experienced unsteady state as shown in **Fig. S3**. The regeneration air temperature gradually increased to the set value, and the CO_2 concentration difference between inlet and outlet increased as well. Then a dynamic steady state was reached, and the temperature and CO_2 concentration remained basically the same. This indicates that the adsorbent experienced adsorption-desorption cycles stably, and the device continuously removed CO_2 from process air and deliver it to regeneration air. By comparing the data change under different conditions, the operation pattern of device can be elucidated.

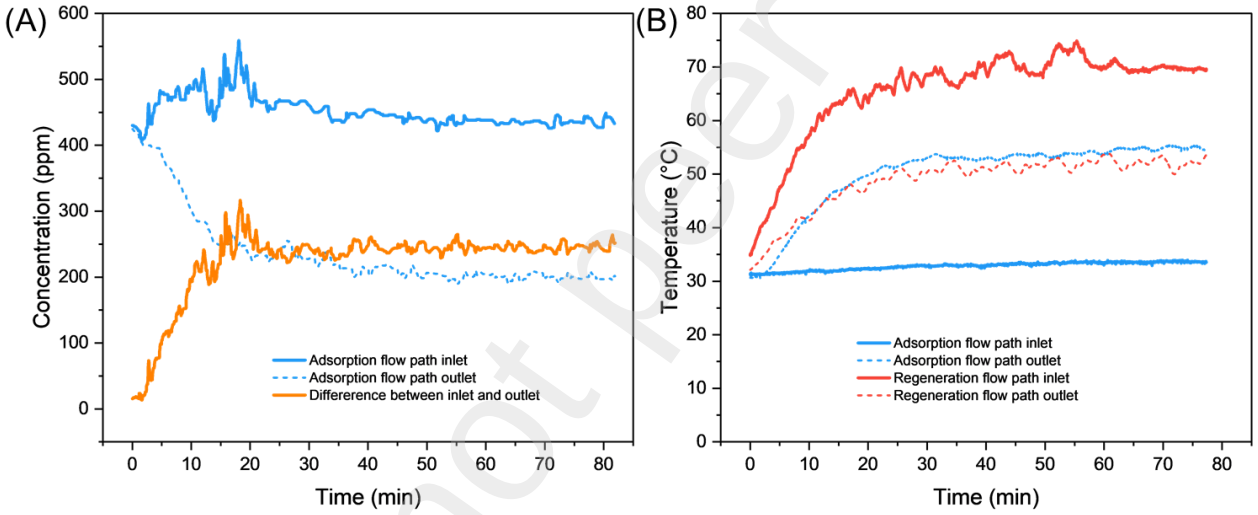


Fig. S3 The unsteady state of operation.

(A) The CO_2 concentration profile.

(B) The temperature profile.

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Device performance indicator calculation

Using the temperature, CO_2 concentration, air flow rate data, the performance indicators of the device are calculated as follows. The carbon capture rate (CCR) is calculated by

$$CCR = \frac{\int_0^t (C_{in,ads} - C_{out,ads})\,dt}{\int_0^t C_{in,ads}\,dt} \times 100\% \quad \text{(S2)}$$

Where $C_{in,ads}$ and $C_{out,ads}$ are inlet and outlet CO_2 concentration (ppm) of adsorption flow path respectively, t is the device test time (min).

The product stream average CO_2 concentration (PC) is calculated by

$$PC = \frac{\int_0^t C_{out,reg}\,dt}{t} \quad \text{(S3)}$$

where $C_{out,reg}$ is the outlet CO_2 concentration (ppm) of regeneration flow path.

The volumetric productivity (Pr_{vol}) is calculated by

$$Pr_{vol} = \frac{\int_0^t (C_{out,reg} - C_{in,reg})\dot{Q}_{reg}\,dt}{V_{vessel}\,t} \quad \text{(S4)}$$

where $C_{in,reg}$ is the inlet CO_2 concentration (ppm) of regeneration flow path, $\dot{Q}_{reg}$ is the regeneration air flow rate ($m^3\ h^{-1}$), V_{vessel} is the volume (m^3) of the adsorber vessel.

The mass productivity (Pr_{mass}) is calculated by

$$Pr_{mass} = \frac{\int_0^t (C_{out,reg} - C_{in,reg})\dot{Q}_{reg}\,dt}{m_{ads}\,t} \quad \text{(S5)}$$

where m_{ads} is the structured adsorbent mass (g).

The energy consumption (EC) is calculated by

$$EC = E_{thermal} + E_{fan} + E_{motor} = \frac{\int_0^t (T_{in,reg} - T_{out,reg})\rho_{air}\dot{Q}_{reg}c_{p,air}\,dt}{60} + \frac{8L\mu_{air}v_{reg}}{R_{channel}^2}\dot{Q}_{reg}t + \frac{8L\mu_{air}v_{pro}}{R_{channel}^2}\dot{Q}_{pro}t + \frac{2\pi n_{rot}T_q}{60}t \quad \text{(S6)}$$

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where $E_{thermal}$ is the thermal energy consumption (kWh),[28] E_{fan} is the fan electricity consumption (kWh),[28] and E_{motor} is the motor electricity consumption (kWh) respectively;[29] $T_{in,reg}$, $T_{out,\ reg}$ are the inlet and outlet temperature (°C) of regeneration flow path respectively; ρ_{air} is the density of air (kg m^{-3}), $c_{p,air}$ is the specific heat capacity of air (kWh kg^{-1} $°C^{-1}$); the coefficient 60 converts the unit min to h; L is the channel length (m), μ_{air} is the viscosity of air (Pa s), v_{reg} is the regeneration air velocity (m s^{-1}), $R_{channel}$ is the channel equivalent diameter (m); n_{rot} is the wheel rotation speed (round per minute), $T_q=mgr\kappa$ is the load torque (N m), g is the gravity (m s^{-2}), r is the wheel radius (m), κ is the dynamic friction assumed to be unity.

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Experimental parametric analysis

The effect of wheel rotation speed is shown in **Fig. S4**.

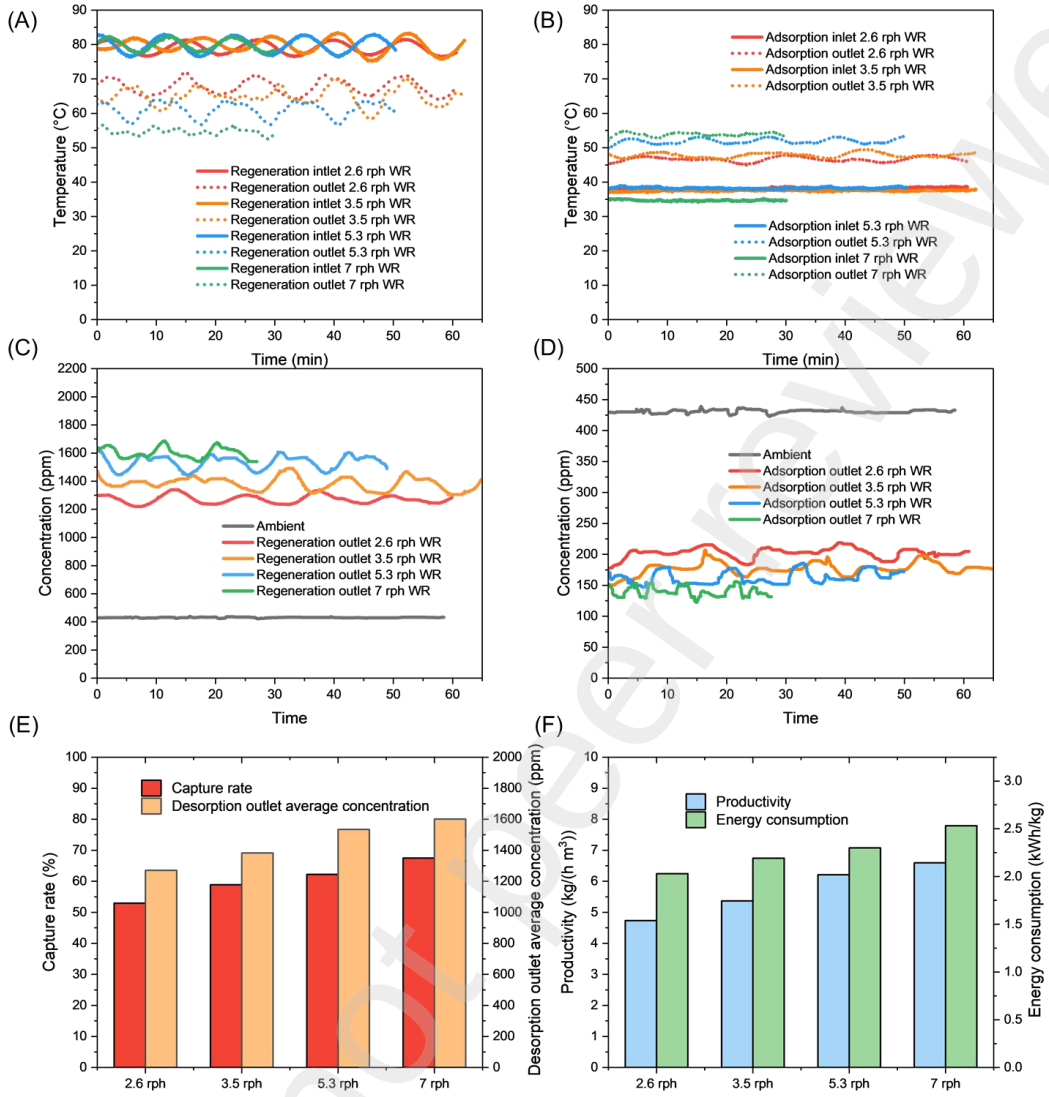


Fig. S4 The effect of wheel rotation speed.

(A) The temperature in inlet and outlet of regeneration flow path.

(B) The temperature in inlet and outlet of adsorption flow path.

(C) The CO_2 concentration in ambience and outlet of regeneration flow path.

(D) The CO_2 concentration in ambience and outlet of adsorption flow path.

(E) The carbon capture rate and product stream CO_2 concentration of device.

(F) The productivity and energy consumption of device.

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The effect of process air flow rate is shown in **Fig. S5**.

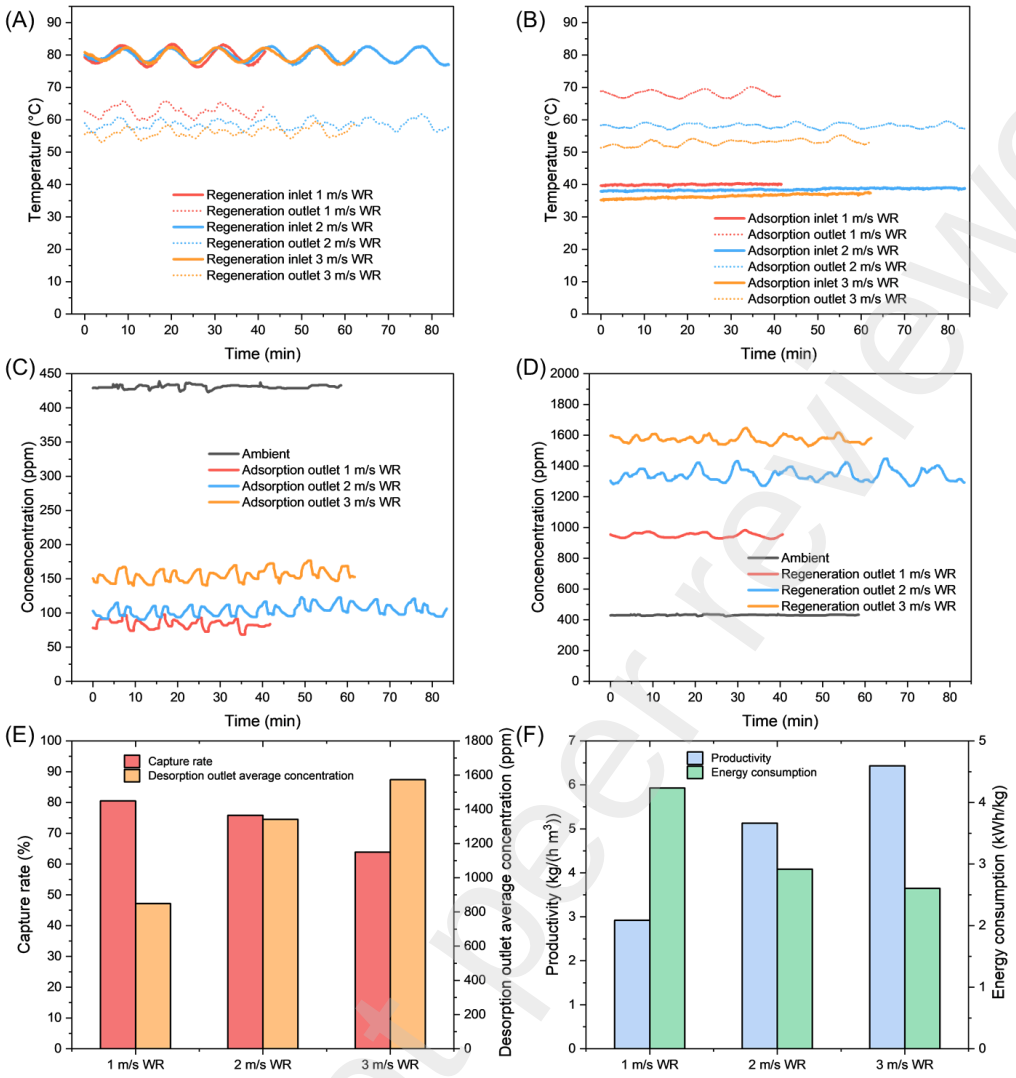


Fig. S5 The effect of process air flow rate

(A) The temperature in inlet and outlet of regeneration flow path.

(B) The temperature in inlet and outlet of adsorption flow path.

(C) The CO_2 concentration in ambience and outlet of adsorption flow path.

(D) The CO_2 concentration in ambience and outlet of regeneration flow path.

(E) The carbon capture rate and product stream CO_2 concentration of device.

(F) The productivity and energy consumption of device.

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Simulation section

Model development

A mathematical model is developed to examine the heat and mass transfer in the temperature swing process. The wheel is divided into adsorption section and regeneration section by sealing partitions. The cross section of channels on the wheel is sinusoidal. Its half width, half height and length are a, b, and L, respectively. During the operation of device, the regeneration air and the process air are arranged in countercurrent, and they enter the channel through the regeneration section and the process section respectively. For a single channel, the process is considered as a one-dimensional axial heat and mass transfer problem. The one-dimensional flow channel is divided into n sub-regions. The sub-region is defined as the control volume, and each of them has a node in its center. The wheel is in direct contact with ambient air, so the boundary conditions are dependent on the inlet airflow properties.

Assumptions

Considering both practical conditions and model complexity, the following assumptions were made.

- All the channels are consistent, and the adsorbents are uniformly distributed on the substrate.
- Ideal gas law and ideal mixtures are assumed for the components, and physical properties of substrate, adsorbent, and gas are all constant.
- The adsorption in the substrate is neglected.
- The temperature and concentration gradients in the radial and circumferential directions of the gas are neglected.
- The axial heat and mass diffusion, axial loss of velocity and pressure drop are neglected.
- The heat and mass diffusion coefficient are constant.
- The local heat balance model is adopted, which means that the temperature of gas in the adsorbent pores is the same as that of the adsorbent.
- The effect of H_2O adsorption is neglected, and other components in air are regarded to be inert.

Control equations

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According to the heat and mass transfer pattern, the control equations are listed as follows.

- Mass conservation in the gas side

$$fA\left(\frac{\partial C_{CO_2,a}}{\partial t}+u\frac{\partial C_{CO_2,a}}{\partial z}\right)=-k_{CO_2}P\left(C_{CO_2,a}-C_{CO_2,s}\right) \tag{S7}$$

where f is the ratio of the air flow channel cross-sectional area to the entire control volume cross-sectional area; A is the control volume cross-sectional area, m^2; $C_{CO_2,a}$ is the molar concentration of CO_2 in the gas phase, mol m^{-3}; $C_{CO_2,s}$ is the molar concentration of CO_2 in the solid phase, mol m^{-3}; t is the time, s; u is the bulk velocity of air, m s^{-1}; z is the spatial coordinate of the axis, m; k_{CO_2} is the convective mass transfer coefficient of CO_2 between the gas-solid side, m s^{-1}; P is the heat transfer perimeter of the channel, m.

- Mass conservation in the solid side

$$\begin{aligned}&\varepsilon(1-f)A\frac{\partial C_{CO_2,a}}{\partial t}+(1-f)A\rho_{ad}\frac{\partial q_{CO_2}}{\partial t}\\&=\varepsilon(1-f)AD_P\frac{\partial^2 C_{CO_2,s}}{\partial z^2}+(1-f)A\rho_{ad}D_S\frac{\partial^2 q_{CO_2}}{\partial z^2}+k_{CO_2}P\left(C_{CO_2,a}-C_{CO_2,s}\right)\end{aligned} \tag{S8}$$

where ε is the adsorbent porosity; ρ_{ad} is the adsorbent density, kg m^{-3}; q_{CO_2} is the CO_2 uptake of adsorbent, mmol g^{-1}; D_P is the intrapore diffusion coefficient of the adsorbent, m^2 s^{-1}; D_S is the surface diffusion coefficient of the adsorbate, m^2 s^{-1}.

- Energy conservation in the gas side

$$\begin{aligned}&\left(c_{p,CO_2}M_{CO_2}C_{CO_2,a}+c_{p,air}\rho_{air}\right)fA\left(\frac{\partial T_a}{\partial t}+u\frac{\partial T_a}{\partial z}\right)\\&=-hP\left(T_a-T_s\right)-c_{p,CO_2}M_{CO_2}k_{CO_2}P\left(C_{CO_2,a}-C_{CO_2,s}\right)\left(T_a-T_s\right)\end{aligned} \tag{S9}$$

where c_{p,CO_2} and $c_{p,air}$ is the constant pressure heat capacity of CO_2 and air respectively, J kg^{-1} K^{-1}; M_{CO_2} is the molar mass of CO_2, kg mol^{-1}; ρ_{air} is the density of air, kg m^{-3}; T_a is the temperature in the gas side, K; T_s is the temperature in the solid side, K; h is the convection heat transfer coefficient between the gas and solid side, W m^{-2} K^{-1}.

- Energy conservation in the solid side

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$$(c_{p,m}l_m + c_{p,ad}l_{ad} + c_{p,CO_2}l_{ad}M_{CO_2}q_{CO_2} + c_{p,CO_2}\varepsilon(1-f)M_{CO_2}C_{CO_2,s}A)\frac{\partial T_s}{\partial t} - (1-f)(1-\varepsilon)A\lambda_{ad}\frac{\partial^2 T_s}{\partial z^2}$$
$$= hP(T_a - T_s) + c_{p,CO_2}M_{CO_2}k_{CO_2}P(C_{CO_2,a} - C_{CO_2,s})(T_a - T_s) + l_{ad}\left(-\Delta H_{CO_2}\frac{\partial q_{CO_2}}{\partial t}\right) \quad \text{(S10)}$$

where $c_{p,m}$ is the heat capacity of the substrate, J kg^{-1} K^{-1}; l_m is the substate mass per unit length, kg m^{-1}; l_{ad} is the adsorbent mass per unit length, kg m^{-1}; λ_{ad} is the heat conduct coefficient of the adsorbent, W m^{-1} K^{-1}; ΔH_{CO_2} is the adsorption heat of CO_2, J mol^{-1}.

Solving conditions

Assuming that the solving starts at the time when the adsorbent enters the regeneration zone, the initial conditions are

$$\begin{cases} T_a(z,0) = T_{a,0} \\ T_s(z,0) = T_{s,0} \\ C_{CO_2,a}(z,0) = C_{CO_2,a,0} \\ C_{CO_2,s}(z,0) = C_{CO_2,s,0} \\ q_{CO_2}(z,0) = q_{CO_2,0} \end{cases} \quad \text{(S11)}$$

where $T_{a,0}$ and $T_{s,0}$ are the initial temperature in the gas and solid phase respectively, $C_{CO_2,a,0}$ and $C_{CO_2,s,0}$ are the initial CO_2 concentration in the gas and solid phase respectively, and $q_{CO_2,0}$ is the initial CO_2 uptake of adsorbent.

The boundary conditions are

$$T_a(0,t) = \begin{cases} T_{r,in}, \text{ regeneration section} \\ T_{p,in}, \text{ process section} \end{cases}$$
$$C_{CO_2,a}(0,t) = \begin{cases} C_{CO_2,r,in}, \text{ regeneration section} \\ C_{CO_2,p,in}, \text{ process section} \end{cases} \quad \text{(S12)}$$

where $T_{r,in}$ and $T_{p,in}$ are the inlet gas temperature in the regeneration and process section respectively, $C_{CO_2,r,in}$ and $C_{CO_2,p,in}$ are the inlet gas CO_2 concentration in the regeneration and process section respectively.

The pseudo second order (PSO) model is used to describe the adsorption kinetics of the adsorbent:

$$\frac{\partial q_{CO_2}}{\partial t} = k_{ad}(q_{e,CO_2} - q_{CO_2})^2 \quad \text{(S13)}$$

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where q_{e,CO_2} is the equilibrium adsorption capacity, mmol g^{-1}; k_{ad} is the rate constant, g $mmol^{-1}$ min^{-1}.

The adsorption isotherms of CO_2 are fitted using the dual site Langmuir-Freundlich model:

$$q_{CO_2} = \frac{q_{e,A} b_A p^{\alpha_A}}{1 + b_A p^{\alpha_A}} + \frac{q_{e,B} b_B p^{\alpha_B}}{1 + b_B p^{\alpha_B}} \tag{S14}$$

where $q_{e,A}$, b_A, α_A, $q_{e,B}$, b_B, α_B are the parameters of dual site Langmuir-Freundlich model.

The isosteric adsorption heat of CO_2 is calculated with Clausius-Clapeyron equation:

$$\Delta H_{CO_2} = -R\left(\frac{\partial \ln p}{\partial (1/T)}\right)_{\theta = const} \tag{S15}$$

The physical properties of the adsorbent are calculated by

$$c_{p,ad} = \frac{((1-\varphi)\rho_{sg} c_{p,sg} + \varphi \rho_{PEI} c_{p,PEI})}{(1-\varepsilon)\rho_{ad}} \tag{S16}$$

$$\lambda_{ad} = \lambda_{PEI} \frac{m_{PEI}}{m_{PEI} + m_{sg}} + \lambda_{sg} \frac{m_{sg}}{m_{PEI} + m_{sg}} \tag{S17}$$

where $\varphi = \frac{V_{PEI}}{V_{ad}} = \frac{V_{PEI}}{V_{PEI} + V_{sg}}$ is the ratio of PEI volume to the adsorbent volume; ρ_{sg} is the silica gel density; ρ_{PEI} is the PEI density; λ_{sg} is the thermal conduct coefficient of silica gel; λ_{PEI} is the thermal conduct coefficient of PEI; $\alpha = \frac{m_{ad}}{m_m} = \frac{m_{PEI} + m_{sg}}{m_m}$ is the ratio of adsorbent mass to the matrix mass; $\beta = \frac{m_{PEI}}{m_{sg}}$ is the ratio of PEI mass to the silica gel mass.

The convection heat transfer coefficient h is calculated by

$$h = \frac{Nu \lambda_{air} P}{4A}$$

where Nu is the Nusselt number; λ_{air} is the thermal conduct coefficient of air, W m^{-1} K^{-1}; $4A/P$ is the hydraulic diameter of channel, m; P is the channel perimeter, m.

The convection mass transfer coefficient of CO_2 k_{CO_2} is calculated by

$$k_{CO_2} = \frac{Sh D_M P}{4A}$$

where Sh is the Sherwood number; D_M is the molecular diffusion coefficient, m^2 s^{-1}. In this case, the Lewis number is close to 1, so Sh=Nu.

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The intrapore diffusion coefficient D_P is calculated by

$$\frac{1}{D_P} = \tau\left(\frac{1}{D_M} + \frac{1}{D_K}\right)$$

where D_K is the Knudsen diffusion coefficient, $m^2\ s^{-1}$; τ is the tortuosity factor of the adsorbent.

The surface diffusion coefficient is calculated by

$$D_S = 1.6 \times 10^{-2} \exp\left(-\frac{0.45 Q_d}{m R_g T}\right)$$

where R_g is the universal gas constant, $J\ mol^{-1}\ K^{-1}$; m is the relation constant.

Model solution

The above control equations are discretized for numerical solution using the backward finite difference method (BFDM). The one-dimensional flow channel is divided into 20 sub-regions. For the 18 internal nodes, the discrete form of equations is as follows.

$$\frac{C^n_{CO_2,a,m} - C^{n-1}_{CO_2,a,m}}{\Delta t} + u\frac{C^n_{CO_2,a,m} - C^n_{CO_2,a,m-1}}{\Delta z} = -w_1\left(C^n_{CO_2,a,m} - C^n_{CO_2,s,m}\right) \tag{S18}$$

$$\begin{aligned}&\frac{C^n_{CO_2,s,m} - C^{n-1}_{CO_2,s,m}}{\Delta t} + w_2\frac{q^n_{CO_2,m} - q^{n-1}_{CO_2,m}}{\Delta t} \\ &= D_p\frac{C^n_{CO_2,s,m+1} - 2C^n_{CO_2,s,m} + C^n_{CO_2,s,m-1}}{\Delta z^2} + w_2 D_s\frac{q^n_{CO_2,m+1} - 2q^n_{CO_2,m} + q^n_{CO_2,m-1}}{\Delta z^2} + w_3\left(C^n_{CO_2,a,m} - C^n_{CO_2,s,m}\right)\end{aligned} \tag{S19}$$

$$\frac{T^n_{a,m} - T^{n-1}_{a,m}}{\Delta t} + u\frac{T^n_{a,m} - T^n_{a,m-1}}{\Delta z} = -w_4^{n-1}\left(T^n_{a,m} - T^n_{s,m}\right) - w_5^{n-1}\left(T^n_{a,m} - T^n_{s,m}\right) \tag{S20}$$

$$\begin{aligned}&w_6^{n-1}\frac{T^n_{s,m} - T^{n-1}_{s,m}}{\Delta t} - w_7^{n-1}\frac{T^n_{s,m+1} - 2T^n_{s,m} + T^n_{s,m-1}}{\Delta z^2} \\ &= hP\left(T^n_{a,m} - T^n_{s,m}\right) + \sum_i^j c_{p,i} M_i k_i P\left(C^{n-1}_{i,a,m} - C^{n-1}_{i,s,m}\right)\left(T^n_{a,m} - T^n_{s,m}\right) + l_{ad}\sum_i^j\left(-\Delta H_i\frac{q^n_{i,m} - q^{n-1}_{i,m}}{\Delta t}\right)\end{aligned} \tag{S21}$$

For the inlet node, the discrete form of equations is modified and listed below.

$$\frac{C^n_{CO_2,a,m} - C^{n-1}_{CO_2,a,m}}{\Delta t} + u\frac{C^n_{CO_2,a,m} - C^n_{CO_2,a,in}}{\Delta z} = -w_1\left(C^n_{CO_2,a,m} - C^n_{CO_2,s,m}\right) \tag{S22}$$

$$\begin{aligned}&\frac{C^n_{CO_2,s,m} - C^{n-1}_{CO_2,s,m}}{\Delta t} + w_2\frac{q^n_{CO_2,m} - q^{n-1}_{CO_2,m}}{\Delta t} \\ &= D_p\frac{C^n_{CO_2,s,m+1} - C^n_{CO_2,s,m}}{\Delta z^2} + w_2 D_s\frac{q^n_{CO_2,m+1} - q^n_{CO_2,m}}{\Delta z^2} + w_3\left(C^n_{CO_2,a,m} - C^n_{CO_2,s,m}\right)\end{aligned} \tag{S23}$$

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$$\frac{T_{a,m}^{n}-T_{a,m}^{n-1}}{\Delta t}+u\frac{T_{a,m}^{n}-T_{a,in}^{n}}{\Delta z}=-w_{4}^{n-1}(T_{a,m}^{n}-T_{s,m}^{n})-w_{5}^{n-1}(T_{a,m}^{n}-T_{s,m}^{n}) \tag{S24}$$

$$\begin{aligned}&w_{6}^{n-1}\frac{T_{s,m}^{n}-T_{s,m}^{n-1}}{\Delta t}-w_{7}^{n-1}\frac{T_{s,m+1}^{n}-T_{s,m}^{n}}{\Delta z^{2}}\\&=hP(T_{a,m}^{n}-T_{s,m}^{n})+\sum_{i}^{j}c_{p,i}M_{i}k_{i}P(C_{i,a,m}^{n-1}-C_{i,s,m}^{n-1})(T_{a,m}^{n}-T_{s,m}^{n})+l_{ad}\sum_{i}^{j}\left(-\Delta H_{i}\frac{q_{i,m}^{n}-q_{i,m}^{n-1}}{\Delta t}\right)\end{aligned} \tag{S25}$$

For the outlet node, the discrete form of equations is modified and listed below.

$$\frac{C_{CO_2,a,m}^{n}-C_{CO_2,a,m}^{n-1}}{\Delta t}+u\frac{C_{CO_2,a,m}^{n}-C_{CO_2,a,m-1}^{n}}{\Delta z}=-w_{1}(C_{CO_2,a,m}^{n}-C_{CO_2,s,m}^{n}) \tag{S26}$$

$$\begin{aligned}&\frac{C_{CO_2,s,m}^{n}-C_{CO_2,s,m}^{n-1}}{\Delta t}+w_{2}\frac{q_{CO_2,m}^{n}-q_{CO_2,m}^{n-1}}{\Delta t}\\&=D_{p}\frac{-C_{CO_2,s,m}^{n}+C_{CO_2,s,m-1}^{n}}{\Delta z^{2}}+w_{2}D_{s}\frac{-q_{CO_2,m}^{n}+q_{CO_2,m-1}^{n}}{\Delta z^{2}}+w_{3}(C_{CO_2,a,m}^{n}-C_{CO_2,s,m}^{n})\end{aligned} \tag{S27}$$

$$\frac{T_{a,m}^{n}-T_{a,m}^{n-1}}{\Delta t}+u\frac{T_{a,m}^{n}-T_{a,m-1}^{n}}{\Delta z}=-w_{4}^{n-1}(T_{a,m}^{n}-T_{s,m}^{n})-w_{5}^{n-1}(T_{a,m}^{n}-T_{s,m}^{n}) \tag{S28}$$

$$\begin{aligned}&w_{6}^{n-1}\frac{T_{s,m}^{n}-T_{s,m}^{n-1}}{\Delta t}-w_{7}^{n-1}\frac{-T_{s,m}^{n}+T_{s,m-1}^{n}}{\Delta z^{2}}\\&=hP(T_{a,m}^{n}-T_{s,m}^{n})+\sum_{i}^{j}c_{p,i}M_{i}k_{i}P(C_{i,a,m}^{n-1}-C_{i,s,m}^{n-1})(T_{a,m}^{n}-T_{s,m}^{n})+l_{ad}\sum_{i}^{j}\left(-\Delta H_{i}\frac{q_{i,m}^{n}-q_{i,m}^{n-1}}{\Delta t}\right)\end{aligned} \tag{S29}$$

For each node, the kinetic models are discretized:

$$\frac{q_{CO_2,m}^{n}-q_{CO_2,m}^{n-1}}{\Delta t}=k(q_{CO_2,e,m}^{n-1}-q_{CO_2,m}^{n-1})^{2} \tag{S30}$$

In this case, there are five equations for each node. For the channel control volume, there are 20 nodes. Thus there are 100 unknown quantities ($C_{CO_2,a}$, $C_{CO_2,s}$, T_a, T_s, q_{CO_2}) and 100 equations in total, which makes up a 100×100 coefficient matrix. The program is compiled with C++ to solve the coefficient matrix of linear equations (the solution process is shown in **Fig. S6**).

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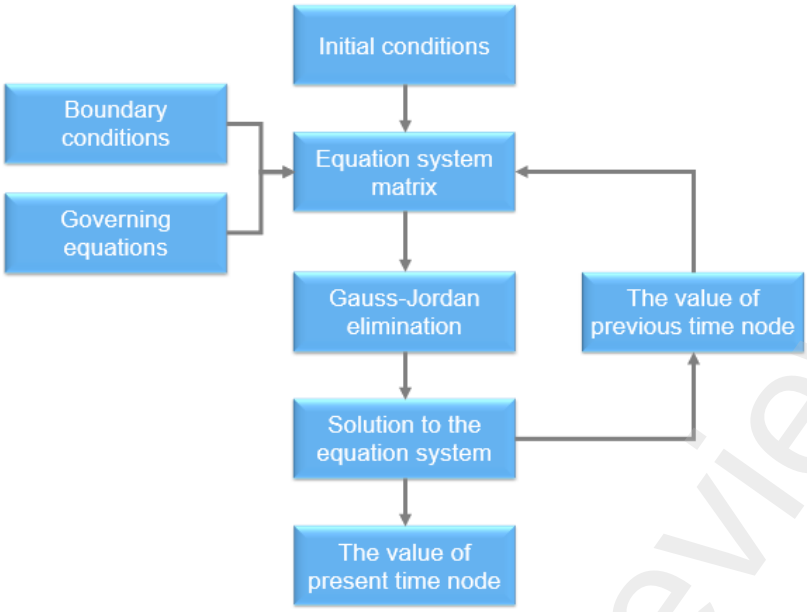


Fig. S6 The solution process of the model.

The parameters used in simulation are shown in **Table S4** and **Table S5**.

Table S4 The main parameters of the rotary adsorption device in the model.

Symbol	Parameter	Value
a	Half channel height	1.5 mm
b	Half channel width	1 mm
L	Wheel thickness	100 mm
θ_p	Process section area fraction	2/3
θ_r	Desorption section area fraction	1/3
$T_{p,\,in}$	Process air inlet temperature	30°C
$C_{CO_2,p,in}$	Process air inlet CO_2 concentration	430 ppm
u_p	Process air velocity	3 m s^{-1}
$T_{r,\,in}$	Regeneration air inlet temperature	60–100°C
$C_{CO_2,r,in}$	Regeneration air inlet CO_2 concentration	430 ppm
u_r	Regeneration air velocity	2 m s^{-1}
ω	Wheel rotation speed	2–7 rph

Table S5 The additional conditions for solving equations.

Symbol	Parameter description	Value
ε	Porosity of solid	0.033
l_m	The mass of substrate in a single channel per unit length	2.104×10^{-3} kg m^{-1}
l_{ad}	The mass of adsorbent in a single channel per unit length	1.896×10^{-3} kg m^{-1}
D_M	Molecular diffusion coefficient of CO_2	1.6×10^{-5} m^2 s^{-1}
D_K	Knudsen diffusion coefficient of CO_2	1.528×10^{-6} m^2 s^{-1}
D_S	Surface diffusion coefficient	9.18×10^{-6} m^2 s^{-1}
f	The cross-sectional area ratio of air flow channel to entire control volume	0.8
ρ_{air}	Air density	1.168 kg m^{-3}
λ_{air}	Air heat conduct coefficient	0.0321 W m^{-1} K^{-1}
λ_{ad}	Adsorbent heat conduct coefficient	0.21 W m^{-1} K^{-1}

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$c_{p,\,air}$	Specific heat capacity at constant pressure of air	1009 J kg^{-1} K^{-1}
c_{p,CO_2}	Specific heat capacity at constant pressure of CO_2	846 J kg^{-1} K^{-1}
$c_{p,\,ad}$	Specific heat capacity at constant pressure of adsorbent	960 J kg^{-1} K^{-1}
$c_{p,\,sub}$	Specific heat capacity at constant pressure of substrate	926 J kg^{-1} K^{-1}

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Simulation parametric analysis

As shown in **Fig. S7**, the wheel thickness was set to 0.05, 0.1, 0.2, 0.3, and 0.4 m, respectively. **Fig. S7** (A) presents that with the thickness increase, the desorbed CO_2 concentration increased while the desorption rate became slower. **Fig. S7** (B) shows that with the thickness increase, the temperature variation rate slowed down. Especially for the 0.4 m wheels, the outlet temperature in regeneration section was always 5°C lower than the inlet temperature, indicating that the adsorbent could not work at suitable temperatures. This is reflected in the gradually reduced working capacity of adsorbent as shown in **Fig. S7** (C-D). **Fig. S7** (E) shows that with the thickness increase, the capture rate increased first and then decreased, while the product stream concentration kept increasing. **Fig. S7** (F) shows that with the thickness increase, the productivity increased first and then decreased, while the energy consumption kept increasing. Above results imply that though thicker wheel contains more adsorbent that is potential to capture more CO2, this will also result in harder heat and mass transfer, so the adsorbent cannot be utilized efficiently. Therefore, the optimized wheel thickness is 0.2–0.3 m in this context.

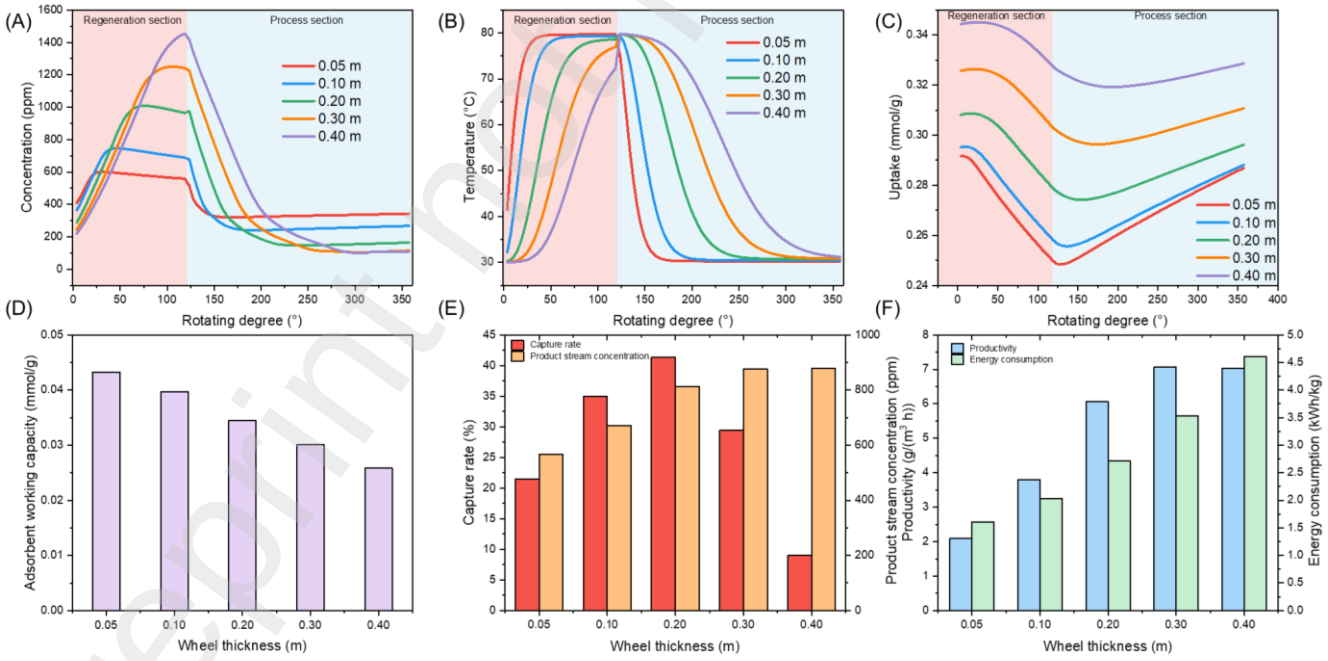


Fig. S7 The effect of wheel thickness.

(A) The CO_2 concentration profile on the wheel outlet surface.

(B) The temperature profile on the wheel outlet surface.

(C) The adsorbent uptake on the wheel outlet surface.

(D) The adsorbent working capacity.

(E) The carbon capture rate and product stream CO_2 concentration.

(F) The productivity and energy consumption.

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As shown in **Fig. S8**, the regeneration section area ratio (*RSAR*) was set to 1/5, 1/4, 1/3, and 1/2, respectively. **Fig. S8** (A) shows that with the *RSAR* increase, the released CO_2 concentration decreased. This is because when air velocity was constant and regeneration area increased, the regeneration air flow rate increased and CO_2 was diluted. **Fig. S8** (B) shows that with the *RSAR* increase, there was more regeneration air whose outlet temperatures reached 80°C. Increasing *RSAR* means extending the regeneration time. **Fig. S8** (C-D) shows that the working capacity increased first then decreased. **Fig. S8** (E) presents that with the *RSAR* increase, capture rate increased and product stream concentration decreased. **Fig. S8** (F) shows that productivity increased first then decreased, and energy consumption had the opposite pattern. Considering working capacity, productivity, and energy consumption, the optimized *RSAR* is around 1/3.

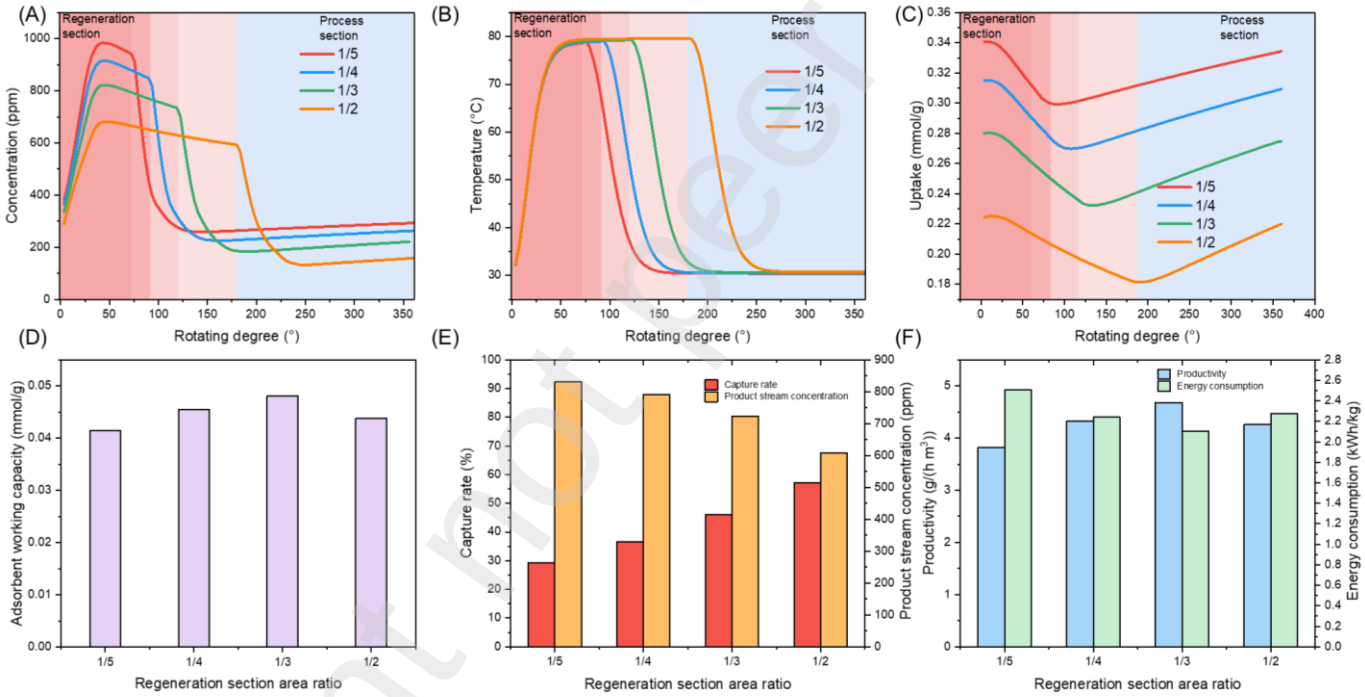


Fig. S8 The effect of regeneration section area ratio.

(A) The CO_2 concentration profile on the wheel outlet surface.

(B) The temperature profile on the wheel outlet surface.

(C) The adsorbent uptake on the wheel outlet surface.

(D) The adsorbent working capacity.

(E) The carbon capture rate and product stream CO_2 concentration.

(F) The productivity and energy consumption.

The effect of channel size is shown in **Fig. S9**. The half channel height *a* was varied from 0.0008 m to 0.0015 m, and half channel width *b* was varied from 0.0012 m to 0.00225 m. **Fig. S9** (A) shows that with the

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increase in channel size, the concentration fluctuation range became smaller. This is possibly because the larger channel size leads to lower adsorbent loading per area, and less CO_2 was captured by the adsorber. **Fig. S9** (B) shows that with the increase in channel size, the temperature change rate increased. The larger channel facilitate the air diffusion, and there is less adsorbent loading per area. The heat transfer was enhanced and heat capacity was lowered, so the temperature change more quickly. **Fig. S9** (C-D) shows that with the increase in channel size, the adsorbent working capacity increased. The larger channel size leads to more efficient utilization of the adsorbent, and thus decreased the specific energy consumption (**Fig. S9** (F)). However, with the increase in channel size, the capture rate, product stream CO_2 concentration, and productivity all decreased. This is possibly because of the lower adsorbent loading per unit area(**Fig. S9** (E-F)).

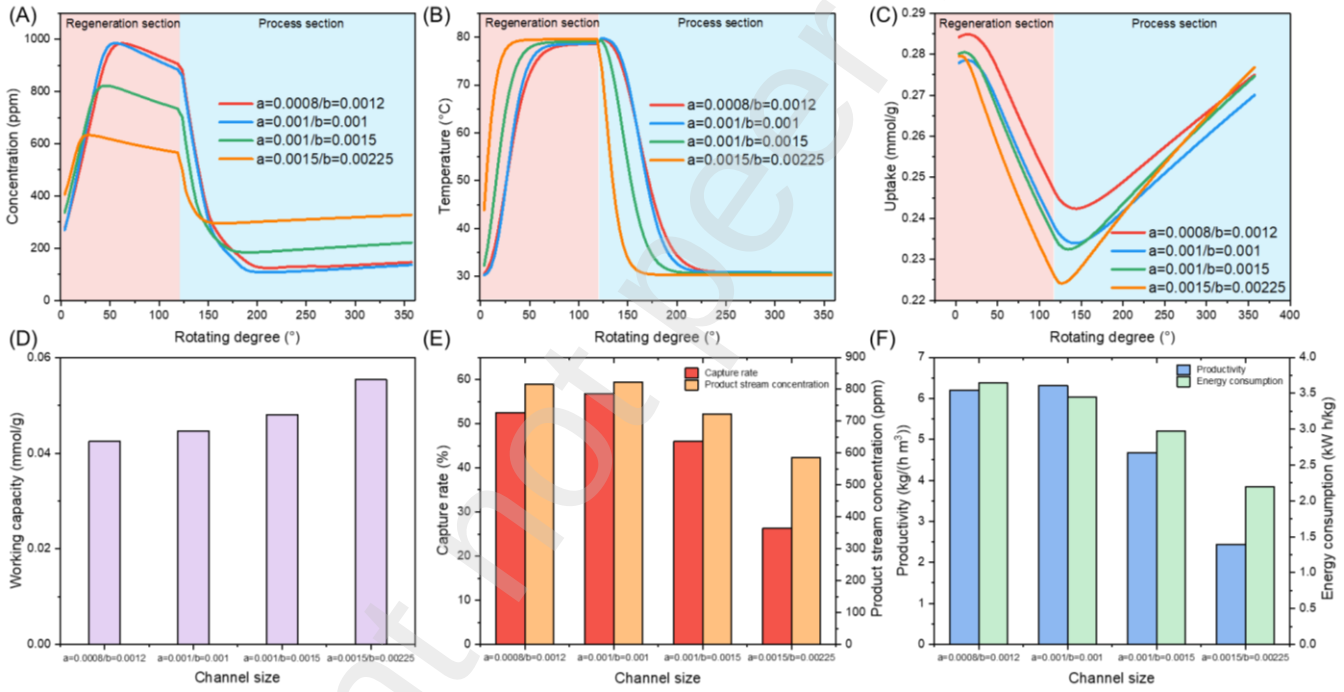


Fig. S9 The effect of channel size.

(A) The CO_2 concentration profile on the wheel outlet surface.

(B) The temperature profile on the wheel outlet surface.

(C) The adsorbent uptake on the wheel outlet surface.

(D) The adsorbent working capacity.

(E) The carbon capture rate and product stream CO_2 concentration.

(F) The productivity and energy consumption.

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Table S6 The orthogonal test table. Nine parameters were investigated and three levels for each parameter were set. The evaluation indicators were carbon capture rate (*CCR*), product stream CO_2 concentration (*PSC*), productivity (*PR*), and energy consumption (*EC*).

No.	l_{ad}	l_m	T_{pro}	v_{pro}	T_{reg}	v_{reg}	ω	L_{wheel}	*RSAR*	*CCR*	*PSC*	*PR*	*EC*	I_{comp}
1	0.0005	0.0005	15	1	60	1	5	0.1	1/4	12.85	615.4	0.896	2.600	0.952
2	0.0005	0.0005	15	1	75	2	7	0.2	1/3	73.58	732.8	4.112	2.325	4.748
3	0.0005	0.0005	15	1	90	3	9	0.3	1/2	60.34	583.4	4.440	4.473	3.450
4	0.0005	0.001	25	2	60	1	5	0.2	1/3	8.48	619.5	1.350	4.229	0.437
5	0.0005	0.001	25	2	75	2	7	0.3	1/2	42.25	656.8	4.985	3.543	3.106
6	0.0005	0.001	25	2	90	3	9	0.1	1/4	15.68	582.2	2.449	3.227	1.252
7	0.0005	0.0015	35	3	60	1	5	0.3	1/2	9.25	584.9	3.076	4.816	0.694
8	0.0005	0.0015	35	3	75	2	7	0.1	1/4	9.02	616.5	2.620	2.919	1.119
9	0.0005	0.0015	35	3	90	3	9	0.2	1/3	16.63	606.0	3.769	5.176	1.134
10	0.001	0.0005	25	3	60	2	9	0.1	1/3	5.80	570.5	1.973	2.524	0.855
11	0.001	0.0005	25	3	75	3	5	0.2	1/2	49.70	653.7	7.360	2.169	4.391
12	0.001	0.0005	25	3	90	1	7	0.3	1/4	10.26	1187.4	4.210	4.267	1.984
13	0.001	0.001	35	1	60	2	9	0.2	1/2	3.86	551.3	2.574	3.839	0.539
14	0.001	0.001	35	1	75	3	5	0.3	1/4	1.93	701.7	4.491	4.730	0.890
15	0.001	0.001	35	1	90	1	7	0.1	1/3	33.89	850.3	3.060	2.904	2.684
16	0.001	0.0015	15	2	60	2	9	0.3	1/4	1.09	594.8	1.775	8.784	-1.040
17	0.001	0.0015	15	2	75	3	5	0.1	1/3	36.75	644.9	4.639	2.296	3.085
18	0.001	0.0015	15	2	90	1	7	0.2	1/2	40.02	1042.0	6.866	4.221	3.811
19	0.0015	0.0005	35	2	60	3	7	0.1	1/2	18.86	504.6	2.281	2.300	1.497
20	0.0015	0.0005	35	2	75	1	9	0.2	1/4	3.41	931.3	2.776	4.082	1.08
21	0.0015	0.0005	35	2	90	2	5	0.3	1/3	32.66	1163.3	9.668	2.492	4.789
22	0.0015	0.001	15	3	60	3	7	0.2	1/4	16.49	756.5	5.378	3.237	2.244
23	0.0015	0.001	15	3	75	1	9	0.3	1/3	8.51	987.4	4.029	5.721	1.201
24	0.0015	0.001	15	3	90	2	5	0.1	1/2	47.22	747.8	6.887	2.211	4.279
25	0.0015	0.0015	25	1	60	3	7	0.3	1/3	8.30	622.4	4.256	5.115	0.912
26	0.0015	0.0015	25	1	75	1	9	0.1	1/2	21.64	828.4	4.434	3.441	2.290
27	0.0015	0.0015	25	1	90	2	5	0.2	1/4	3.20	917.4	5.373	3.906	1.677

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Table S7 Analysis of the orthogonal test results. Range analysis method is used to evaluate effects of different parameters when *CCR*, *PSC*, *PR*, *EC*, and comprehensive evaluation index I_{comp} are used as the indicator, respectively.

	l_{ad}	l_m	T_{pro}	v_{pro}	T_{reg}	v_{reg}	ω	L_{wheel}	*RSAR*
Level	Carbon capture rate *CCR* as indicator								
k1	248.08	267.46	296.85	219.59	84.98	148.31	202.04	201.71	73.93
k2	183.30	178.31	165.31	199.20	246.79	218.68	252.67	215.37	224.60
k3	160.29	145.90	129.51	172.88	259.90	224.68	136.96	174.59	293.14
Range	87.79	121.56	167.34	46.71	174.92	76.37	115.71	40.78	219.21
Optimal	0.0005	0.0005	15	2	90	2	7	0.1	1/2
Level	Product stream CO_2 concentration *PSC* as indicator								
k1	5597.5	6942.4	6705	6403.1	5419.9	7646.6	6648.6	5960.6	6903.2
k2	6796.6	6453.5	6638.3	6739.4	6753.5	6551.2	6969.3	6810.5	6797.1
k3	7459.1	6457.3	6509.9	6710.7	7679.8	5655.4	6235.3	7082.1	6152.9
Range	1861.6	488.9	195.1	336.3	2259.9	1991.2	734	1121.5	750.3
Optimal	0.0015	0.0005	15	2	90	1	7	0.3	1/4
Level	Productivity *PR* as indicator								
k1	27.697	37.716	39.022	33.636	23.559	30.697	43.74	29.239	29.968
k2	36.948	35.203	36.390	36.789	39.446	39.967	37.768	39.558	36.856
k3	45.082	36.808	34.315	39.302	46.722	39.063	28.219	40.93	42.903
Range	17.385	2.513	4.707	5.666	23.163	9.27	15.521	11.691	12.935
Optimal	0.0015	0.0005	15	3	90	2	5	0.3	1/2
Level	Energy consumption *EC* as indicator								
k1	33.308	27.232	35.868	33.333	37.444	36.281	29.449	24.422	37.752
k2	35.734	33.641	32.421	35.174	31.226	32.543	30.831	33.184	32.782
k3	32.505	40.674	33.258	33.04	32.877	32.723	41.267	43.941	31.013
Range	3.229	13.442	3.447	2.134	6.218	3.738	11.818	19.519	6.739
Optimal	0.0015	0.0005	25	3	75	3	5	0.1	1/2
Level	Comprehensive evaluation index I_{comp} as indicator								
k1	16.89	23.69	22.73	18.14	7.09	15.07	21.19	18.01	10.10
k2	17.20	16.63	16.91	17.96	21.85	20.07	22.10	20.00	19.85
k3	19.91	13.68	14.36	17.90	25.06	18.86	10.70	15.99	24.06
Range	3.02	10.00	8.37	0.24	17.97	5.00	11.40	4.01	13.96
Optimal	0.0015	0.0005	15	3	90	2	5	0.2	1/2

The comprehensive evaluation index of group No. i is defined as

$$I_{\mathrm{comp},i} = \overline{CCR_i} + \overline{PSC_i} + \overline{PR_i} - \overline{EC_i}$$

where $\overline{CCR}_i$, $\overline{PSC}_i$, $\overline{PR}_i$, $\overline{EC}_i$, are the dimensionless number of CCR_i, PSC_i, PR_i, EC_i respectively, which are determined by

$$\overline{CCR_i} = \frac{CCR_i}{\sum_{1}^{27} CCR_i}$$

$$\overline{PSC_i} = \frac{PSC_i}{\sum_{1}^{27} PSC_i}$$

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$$\overline{PR_i} = \frac{PR_i}{\sum_{1}^{27} PR_i}$$

$$\overline{EC_i} = \frac{EC_i}{\sum_{1}^{27} EC_i}$$

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Techno-economic analysis

The report details specific CAPEX and OPEX for an integrated DAC system comprising a 1000 m^2 greenhouse.

Financial assumptions

- A plant lifetime of 20 years and an interest rate i of 10% to calculate capital recovery factor (CRF) are assumed.
- The rotary adsorber operates during the day (12 hours per day) to supply CO_2 for photosynthesis. The system operates 328 days per year.
- A 5% income tax was assumed for the product selling.
- Other specific assumptions are mentioned in the corresponding calculations.

Capital cost

- Adsorber

The capital cost of rotary adsorber prototype with a capacity of 0.7 tCO_2/year built in this study is \$3011, and it is used as a base case. With the daily productivity of 342 kg day^{-1}, the annual capacity of the adsorber will be 112 tCO_2 $year^{-1}$. Thus, the capital cost of the rotary adsorber is scaled with a scaling exponent of 0.53[30] by

$$Cost_{Adsorber} = 3011 \times \left(\frac{112}{0.7}\right)^{0.53} = \$44413$$

- Heat exchanger

Two heat exchangers (HX) are needed in this system. With the hypothetic air states in different points summarized in **Table S8**, the heat exchanger area is calculated from

$$A_{HX} = \frac{Q_{ex}}{h_{conv}(T_{out} - T_{in})}$$

where Q_{ex} is the heat exchange power (W), T_{in} and T_{out} are the fluid temperature (°C) at inlet and outlet of the

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heat exchanger. Other parameter meaning and values are listed in **Table S11**. The tube-fin heat exchanger is considered, and the cost is scaled using a base 19.13 m^2 heat exchanger with cost of \$383.48 and a scaling exponent of 0.53[30] by

$$Cost_{HX} = 383.48 \times \left(\frac{A_{HX}}{19.13}\right)^{0.53}$$

The calculated area A_{HX1} and A_{HX2} is 99 and 115 m^2 respectively, so $Cost_{HX1}$ is \$916 and $Cost_{HX2}$ is \$994.

- Fan

Two fans are needed in this system. With the hypothetic air states in different points summarized in **Table S8**, the pressure drop required to be overcome by fan power [28] and the fan cost are calculated from[31]

$$\Delta p = \frac{8lv\mu_{air}}{d^2}$$

$$Cost_{fan} = e^{6.8929 + 0.74\ln\left(\frac{\Delta pV_{fan}}{745.7}\right)}$$

where v is Air velocity in channel (m s^{-1}) and V_{fan} is the air volumetric flow rate in the fan (m^3 h^{-1}), other parameter meaning and value are listed in **Table S11**.

Results show that $Cost_{fan1}$ is \$2456 and $Cost_{fan2}$ is \$1218.

The expenditure of installation, piping, electrical facilities, and land are considered among direct costs as a percentage of equipment purchase costs. Engineering and supervision and construction expenses are considered among indirect costs as a percentage of equipment purchase costs. All capital costs are calculated and shown in

Table **S9**.

Table S8 The point status marked in Figure 6.

Point	Status
a	Air, 48600 m^3/h, 410 ppm, 25°C, 50%RH, 50.32 kJ/kg, 9.88 g/kg
b	Air, 48600 m^3/h, 111 ppm, 41°C, 13.4%RH, 58.67 kJ/kg, 6.76 g/kg
c	Air, 18000 m^3/h, 410 ppm, 25°C, 50%RH, 50.32 kJ/kg, 9.88 g/kg
d	Air, 18000 m^3/h, 410 ppm, 55°C, 9.5%RH, 81.92 kJ/kg, 9.88 g/kg
e	Air, 18000 m^3/h, 410 ppm, 80°C, 3%RH, 106.66 kJ/kg, 9.88 g/kg

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f	Air, 18000 m^3/h, 1217 ppm, 60°C, 20%RH, 126.95 kJ/kg, 25.49 g/kg
g	Air, 18000 m^3/h, 1217 ppm, 30°C, 94%RH, 95.35 kJ/kg, 25.49 g/kg
h	Water vapor, 5030 kg/h, 160°C, 2789.57 kJ/kg
i	Water vapor, 5030 kg/h, 110°C, 2683.34 kJ/kg

Table S9 The capital expenditure (CAPEX) of system.

Item	Value
Adsorber	
Capacity (t $year^{-1}$)	112
Cost ($)	44413
HX1	
Power (kW)	148
Area (m^2)	99
Cost ($)	916
HX2	
Power (kW)	208
Area (m^2)	115
Cost ($)	994
Fan 1	
Air flow rate ($m^3\ h^{-1}$)	48600
Cost ($)	2456
Fan 2	
Air flow rate ($m^3\ h^{-1}$)	18000
Cost ($)	1218
Equipment cost ($)	49998
Installation ($)	2500
Piping ($)	500
Electrical ($)	750
Land ($)	1000
Total direct FCI ($)	54747
Engineering and supervision ($)	1250
Construction expenses ($)	2500
Total indirect FCI ($)	3750
Fixed capital investment (FCI) ($)	58497
Interest rate, *i*	0.1
Lifetime (years)	20
Capital recovery factor	0.117
CAPEX ($/year)	6871
CAPEX ($/t)	61

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Operation cost

- Adsorbent cost

The adsorbent price is calculated using the method of Sinha et al.[28] Based on the raw material price of polyethyleneimine, ceramic fiber, and silica sol, the bulk price of the structured adsorbent is estimated to be \$5.37 kg^{-1}. Amine-modified structured adsorbents are assumed to have the lifetime of three years and be replaced after their service life.[32] The adsorbent loading in the rotary adsorber is 2375 kg, then the adsorber cost is \$4250 $year^{-1}$.

- Electric energy consumption cost

The power requirements of fans W_{fan} were calculated as the total of the energy required to push the air through the adsorber and the energy required to push the air through the heat exchanger by

$$W_{fan} = W_{fan,adsorber} + W_{fan,HX}$$

The energy required to push the air through the adsorber is estimated by[31]

$$W_{fan,adsorber} = \frac{V_{fan}\Delta p}{\eta_{fan}}$$

The parameter meaning and value are listed in **Table S11**. The energy required to push the air through the heat exchanger $W_{fan,HX}$ is estimated as 0.005 W per 1 W of heat exchange [33]. The heat exchange power is 148 kW and 208 kW for *HX1* and *HX2* respectively. Thus, the W_{fan1} is 2963 W and W_{fan2} is 2268 W.

The motor power of rotary adsorber is calculated by

$$P_{motor} = T_{motor}\frac{2\pi n_{motor}}{60}$$

The parameter meaning and value are listed in **Table S11**. Results show the motor power P_{motor} is 2529 W.

The electricity is obtained from the grid, solar photovoltaics, hydroelectric, or natural gas with carbon capture and storage (CCS), and their costs are shown in **Table S10**. Depending on the electricity source, the total annual electricity energy consumption cost is \$1527–3054 $year^{-1}$.

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Table S10 The electricity cost of different sources.

Source	Electricity cost ($ kWh^{-1})
Grid	0.1[34]
Solar photovoltaics	0.05[35]
Hydroelectric	0.062[35]
Natural gas with CCS	0.075[35]

- Thermal energy consumption cost

Based on the prototype experiments, thermal energy consumption of the rotary adsorber is assumed to be 2.2 kWh kg^{-1}. According to the adsorber's productivity of 342 kg day^{-1} and a waste heat cost of \$0–0.024 kWh^{-1},[4, 34, 36] the annual thermal energy consumption cost is \$0–5929 year^{-1}.

Maintenance and repairs are evaluated as a percentage of equipment cost. Operating labor, fixed charges (depreciation and insurance), and distribution and marketing costs are evaluated as a percentage of direct production costs. Administrative costs are evaluated as a percentage of operating labor. All operation costs are calculated and shown in **Table S12**.

Table S11 The parameter meaning and value.

Parameter	Meaning	Value
h_{conv}	Convection heat transfer coefficient (W m^{-2} °C^{-1})	60
l	Channel length (m)	1.1
μ_{air}	Air viscosity (kg m^{-1} s^{-1})	1.84×10^{-5}
d	Structured adsorbent channel inner radius (m)	0.0012
η_{fan}	Fan efficiency (%)	80
T_{motor}	Motor torque of the motor (N m)	23000
n_{motor}	Rotor rotation speed of the motor (r min^{-1})	1

Table S12 Operation expenditure (OPEX) of system.

Item	Value
Adsorbent (\$/year)	4250
Electrical energy (\$/year)	1527–3054
Thermal energy (\$/year)	0–5929
Maintenance and repairs (\$/year)	500
Operating labor	687
Direct production costs (\$/year)	14421
Fixed charger (\$/year)	1442
Administrative costs (\$/year)	103

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Distribution and selling costs ($/year)	288
OPEX ($/year)	7429–16254
OPEX ($/t)	66–145

Revenue and capital recovery period

The revenue of the integrated system is the tomato selling profit. The planting density is 3 plant m^{-2},[37] for a 800 m^2 greenhouse, 2400 tomato plants can be grown. Assume that when CO_2 enrichment is performed to create a 1000 ppm CO_2 atmosphere in greenhouse using the rotary adsorber, it takes about 65 days to harvest after sowing (consider the annual 328-day operation, five harvest can be done each year),[38] and the yield increased by 0.9 kg per plant under the effects of CO_2 enrichment.[39] According to a tomato selling price of 3 $ kg^{-1},[40] the extra annual profit when using the strategy proposed by this paper is $15350–24175 $year^{-1}$ as shown in **Table S13**.

In the above calculation, 112 t CO_2 can be captured by the adsorber and transported to the greenhouse with the total capture cost of $127–206 t^{-1}. Given the capital cost of constructing the integrated system of $58497, the capital recovery period can be estimated to be 2.4–3.8 years.

Table S13 The revenue and capital recovery period of CO_2 enrichment.

Item	Value
Greenhouse area (m^2)	1000
Plant density (plant m^{-2})	3
Total plants (plant)	3000
Production per plant without CO_2 enrichment (kg)	3.0
Production per plant with CO_2 enrichment (kg)	3.9
Production increase (kg)	0.9
Tomato selling price ($ kg^{-1})	3
Harvest time per year	5
Revenue of the CO_2 enrichment ($ $year^{-1}$)	40500
Income tax rate (%)	5
Net revenue of the CO_2 enrichment ($ $year^{-1}$)	38475
Total cost of the CO_2 enrichment ($ $year^{-1}$)	14300–23125
Net profit of the CO_2 enrichment ($ $year^{-1}$)	15350–24175
Capital recovery period (year)	2.4–3.8

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Reference

[1] J.A. Wurzbacher, C. Gebald, A. Steinfeld, Separation of CO2 from air by temperature-vacuum swing adsorption using diamine-functionalized silica gel, Energy & Environmental Science 4(9) (2011). https://doi.org/10.1039/c1ee01681d.

[2] J.A. Wurzbacher, C. Gebald, N. Piatkowski, A. Steinfeld, Concurrent separation of CO2 and H2O from air by a temperature-vacuum swing adsorption/desorption cycle, Environ Sci Technol 46(16) (2012) 9191-8. https://doi.org/10.1021/es301953k.

[3] C.J. E. Bajamundi, J. Koponen, V. Ruuskanen, J. Elfving, A. Kosonen, J. Kauppinen, J. Ahola, Capturing CO2 from air: Technical performance and process control improvement, Journal of CO2 Utilization 30 (2019) 232-239. https://doi.org/10.1016/j.jcou.2019.02.002.

[4] M.M. Sadiq, M.P. Batten, X. Mulet, C. Freeman, K. Konstas, J.I. Mardel, J. Tanner, D. Ng, X. Wang, S. Howard, M.R. Hill, A.W. Thornton, A Pilot - Scale Demonstration of Mobile Direct Air Capture Using Metal - Organic Frameworks, Advanced Sustainable Systems 4(12) (2020). https://doi.org/10.1002/adsu.202000101.

[5] H.M. Schellevis, T.N. van Schagen, D.W.F. Brilman, Process optimization of a fixed bed reactor system for direct air capture, International Journal of Greenhouse Gas Control 110 (2021) 103431. https://doi.org/https://doi.org/10.1016/j.ijggc.2021.103431.

[6] W. Zhang, H. Liu, C. Sun, T.C. Drage, C.E. Snape, Capturing CO2 from ambient air using a polyethyleneimine–silica adsorbent in fluidized beds, Chemical Engineering Science 116 (2014) 306-316. https://doi.org/10.1016/j.ces.2014.05.018.

[7] Q. Yu, W. Brilman, A Radial Flow Contactor for Ambient Air CO2 Capture, Applied Sciences 10(3) (2020). https://doi.org/10.3390/app10031080.

[8] C. Tang, X. Gao, Y. Shao, L. Wang, K. Liu, R. Gao, D. Che, Investigation on the rotary regenerative adsorption wheel in a new strategy for CO2 enrichment in greenhouse, Applied Thermal Engineering 205 (2022). https://doi.org/10.1016/j.applthermaleng.2022.118043.

[9] G.R. Majeed S. Jassim, Dag Eimer, and Colin Ramshaw, **Carbon Dioxide Absorption and Desorption in Aqueous Monoethanolamine Solutions in a Rotating Packed Bed**, Ind. Eng. Chem. Res. 46 (2007) 2823-2833.

[10] A.S. Joel, M. Wang, C. Ramshaw, E. Oko, Process analysis of intensified absorber for post-combustion CO2 capture through modelling and simulation, International Journal of Greenhouse Gas Control 21 (2014) 91-100. https://doi.org/10.1016/j.ijggc.2013.12.005.

[11] A.S. Joel, M. Wang, C. Ramshaw, E. Oko, Modelling, simulation and analysis of intensified regenerator for solvent based carbon capture using rotating packed bed technology, Applied Energy 203 (2017) 11-25. https://doi.org/10.1016/j.apenergy.2017.05.157.

[12] I.M. Dahl, E. Bakken, Y. Larring, A.I. Spjelkavik, S.F. Håkonsen, R. Blom, On the development of novel reactor concepts for chemical looping combustion, Energy Procedia 1(1) (2009) 1513-1519. https://doi.org/10.1016/j.egypro.2009.01.198.

[13] Z. Zhao, C.O. Iloeje, T. Chen, A.F. Ghoniem, Design of a rotary reactor for chemical-looping combustion. Part 1: Fundamentals and design methodology, Fuel 121 (2014) 327-343. https://doi.org/10.1016/j.fuel.2013.11.056.

[14] Z. Zhao, A.F. Ghoniem, Design of a rotary reactor for chemical-looping combustion. Part 2: Comparison of copper-, nickel-, and iron-based oxygen carriers, Fuel 121 (2014) 344-360. https://doi.org/10.1016/j.fuel.2013.11.055.

[15] T.S. Ge, Y.J. Dai, R.Z. Wang, Y. Li, Experimental investigation on a one-rotor two-stage rotary desiccant cooling system, Energy 33(12) (2008) 1807-1815. https://doi.org/10.1016/j.energy.2008.08.006.

[16] A.K. Hisashi Yamauchi, Tsutomu Hirose, Hiroshi Okano, and Ken-ichiro Yamada, Performance of VOC Abatement by Thermal Swing Honeycomb Rotor Adsorbers, Ind. Eng. Chem. Res. 46(12) (2007) 4316–4322.

[17] Y. Sheng, L. Zhang, Y. Wang, L. Fang, Explore energy saving operation strategy: Indoor VOCs removal performance of silica gel rotor in clean-air heat pump system at low regeneration air temperature, Energy and Buildings 202 (2019). https://doi.org/10.1016/j.enbuild.2019.109379.

[18] T. Gupta, R. Ghosh, Rotating bed adsorber system for carbon dioxide capture from flue gas, International Journal of Greenhouse Gas Control 32 (2015) 172-188. https://doi.org/10.1016/j.ijggc.2014.10.020.

[19] J.A.A. Gibson, E. Mangano, E. Shiko, A.G. Greenaway, A.V. Gromov, M.M. Lozinska, D. Friedrich, E.E.B. Campbell, P.A. Wright, S. Brandani, Adsorption Materials and Processes for Carbon Capture from Gas-Fired Power Plants: AMPGas, Industrial & Engineering Chemistry Research 55(13) (2016) 3840-3851. https://doi.org/10.1021/acs.iecr.5b05015.

[20] Ghaffari-Nik, Rapid Cycle Temperature Swing Adsorption Process Using Solid Structured Sorbent for CO2 capture from

Preprint not peer reviewed

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Cement Flue Gas, GHGT-15, 2021.
[21] L. Herraiz, E. Palfi, E. Sánchez Fernández, M. Lucquiaud, Rotary Adsorption: Selective Recycling of CO2 in Combined Cycle Gas Turbine Power Plants, Frontiers in Energy Research 8 (2020). https://doi.org/10.3389/fenrg.2020.482708.
[22] L. Wang, Y. Shao, X. Chen, X. Yang, O. Manca, J. Yan, D. Che, Investigation on a novel post-combustion CO2 capture rotary adsorption wheel based on temperature swing adsorption, International Journal of Green Energy (2022) 1-13. https://doi.org/10.1080/15435075.2022.2119857.
[23] J. Gao, Y. Hoshino, G. Inoue, Honeycomb-carbon-fiber-supported amine-containing nanogel particles for CO2 capture using a rotating column TVSA, Chemical Engineering Journal 383 (2020). https://doi.org/10.1016/j.cej.2019.123123.
[24] H. Yang, X. Fang, Z. Li, H. Sun, H. Chen, Copper-Doped Small Pore Zeolites for CO2 Capture by Honeycomb Rotor with Low Temperature Regeneration, ACS Sustainable Chemistry & Engineering 10(5) (2022) 1759-1764. https://doi.org/10.1021/acssuschemeng.1c08347.
[25] K.N. Son, G.E. Cmarik, J.C. Knox, J.A. Weibel, S.V. Garimella, Measurement and Prediction of the Heat of Adsorption and Equilibrium Concentration of CO2 on Zeolite 13X, Journal of Chemical & Engineering Data 63(5) (2018) 1663-1674. https://doi.org/10.1021/acs.jced.8b00019.
[26] T.M. McDonald, W.R. Lee, J.A. Mason, B.M. Wiers, C.S. Hong, J.R. Long, Capture of carbon dioxide from air and flue gas in the alkylamine-appended metal-organic framework mmen-Mg2(dobpdc), J Am Chem Soc 134(16) (2012) 7056-65. https://doi.org/10.1021/ja300034j.
[27] J. Wu, X. Zhu, F. Yang, T. Ge, R. Wang, Easily-synthesized and low-cost amine-functionalized silica sol-coated structured adsorbents for CO2 capture, Chemical Engineering Journal 425 (2021). https://doi.org/10.1016/j.cej.2021.131409.
[28] A. Sinha, L.A. Darunte, C.W. Jones, M.J. Realff, Y. Kawajiri, Systems Design and Economic Analysis of Direct Air Capture of CO2 through Temperature Vacuum Swing Adsorption Using MIL-101(Cr)-PEI-800 and mmen-Mg2(dobpdc) MOF Adsorbents, Industrial & Engineering Chemistry Research 56(3) (2017) 750-764. https://doi.org/10.1021/acs.iecr.6b03887.
[29] J.C. Atuonwu, G. van Straten, H.C. van Deventer, A.J.B. van Boxtel, On the controllability and energy sensitivity of heat-integrated desiccant adsorption dryers, Chemical Engineering Science 80 (2012) 134-147. https://doi.org/10.1016/j.ces.2012.06.006.
[30] R.W. Whitesides, Process Equipment Cost Estimating By Ratio And Proportion, 2020.
[31] N. McQueen, P. Psarras, H. Pilorge, S. Liguori, J. He, M. Yuan, C.M. Woodall, K. Kian, L. Pierpoint, J. Jurewicz, J.M. Lucas, R. Jacobson, N. Deich, J. Wilcox, Cost Analysis of Direct Air Capture and Sequestration Coupled to Low-Carbon Thermal Energy in the United States, Environ Sci Technol 54(12) (2020) 7542-7551. https://doi.org/10.1021/acs.est.0c00476.
[32] H. Azarabadi, K.S. Lackner, A sorbent-focused techno-economic analysis of direct air capture, Applied Energy 250 (2019) 959-975. https://doi.org/10.1016/j.apenergy.2019.04.012.
[33] J.R. Couper, W.R. Penney, J.R. Fair, S.M. Walas, Chemical process equipment: selection and design, Gulf professional publishing2005.
[34] C. Bauer, S. Hirschberg, B. Cox, Y. Baeuerle, X. Zhang, K. Treyer, T. Heck, W. Schenler, S. Biollaz, A. Calbry-Muzyka, M. Lehnert, A. Meier, H.-M. Prasser, F. Vogel, H.-C. Wieckert, M. Zimmermann, V. Burg, M. Saar, G. Bowman, M. Tran, Potentials, costs and environmental assessment of electricity generation technologies, 2017.
[35] U.S.D.o.E.E.I. Administration, Annual Energy Outlook, 2018. https://www.eia.gov/outlooks/archive/aeo18/pdf/electricity_generation.pdf.
[36] G. Leonzio, P.S. Fennell, N. Shah, A Comparative Study of Different Sorbents in the Context of Direct Air Capture (DAC): Evaluation of Key Performance Indicators and Comparisons, Applied Sciences 12(5) (2022). https://doi.org/10.3390/app12052618.
[37] K.E. Cockshull, L.C. Ho, Regulation of tomato fruit size by plant density and truss thinning, Journal of Horticultural Science 70(3) (1995) 395-407. https://doi.org/10.1080/14620316.1995.11515309.
[38] F. Vegetables, HARVESTING TOMATOES MADE EAS, 2016. https://www.espoma.com/fruits-vegetables/harvesting-tomatoes-made-easy/.
[39] S. Nilsen, K. Hovland, C. Dons, S.P. Sletten, Effect of CO2 enrichment on photosynthesis, growth and yield of tomato, Scientia Horticulturae 20(1) (1983) 1-14.
[40] M. Shahbandeh, Retail price of tomatoes (field grown) in the United States from 1995 to 2021, 2022. https://www.statista.com/statistics/236907/retail-price-of-tomatoes-in-the-united-states/.

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